

Social Interaction and Scientists' Productivity: Evidence from Historical Correspondence Networks in Quantum Mechanics, 1919–1932

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How does direct communication shape scientists' responses to a major intellectual breakthrough? I study this question around the emergence of quantum mechanics in 1925. I combine newly digitized records of correspondence among physicists with publication and citation data from the Web of Science, constructing a balanced panel of 1,343 scientists from 1919 to 1932. Treatment is a recorded pre-1925 correspondence link to at least one of eight founders of quantum mechanics. Because connected scientists were more productive before 1925, I supplement two-way fixed effects with inverse-probability-weighted event studies and doubly robust difference-in-differences estimators that condition on pre-discovery productivity and scientist characteristics. The fixed-effects estimates show that, after 1925, connected scientists became 5.4 percentage points more likely to publish and produced work receiving approximately 0.12–0.13 log points more citations. The publication-count estimate is positive but imprecise. Weighted event studies show a sustained positive citation differential, which remains present in theoretical quantum studies but has no comparable sustained pattern in astrophysics and physical chemistry. Correspondence links were not randomly assigned, part of the archival timing overlaps 1925, and indirect knowledge diffusion may contaminate the comparison group. I therefore interpret the estimates as conditional evidence that pre-existing access to the founders facilitated scientific production and recognition during the quantum-mechanics revolution.

Roman Dimakov: Centre for Economic Research and Graduate Education – Economic Institute, a joint workplace of Charles University and the Economics Institute of the Czech Academy of Sciences, Politických veznu 7, 111 21 Prague, Czech Republic. Correspondence e-mail: Roman.Dimakov@cerge-ei.cz. I want to give special thanks to the following people for interacting with me for discussion of my project and listening to my ideas and for allowing me to share with them the fascinating history of quantum physics and physics in general. These people are: Cagin Keskin, Vladimir Shchukin, Samir Orujov, my supervisor Paolo Zacchia, economic historians Sebastian Ottinger and Christian Ochsner, and other faculty members at CERGE-EI.

1. Introduction

How does personal communication affect scientists' ability to respond to a major intellectual breakthrough? Scientific knowledge is made public through articles, books, and conferences, yet researchers also exchange unfinished arguments, technical advice, and early results through personal relationships. Such interactions may be especially valuable when a new theory changes the questions, methods, and language of a field. Scientists with direct access to the researchers developing the theory may learn sooner, resolve uncertainty more quickly, or identify productive applications before otherwise comparable peers. Measuring this advantage is difficult because informal communication is rarely observed and because scientists choose their professional contacts.

This paper studies these questions during the quantum-mechanics revolution of 1925–1927. The setting combines a sharp intellectual break with unusually rich evidence on communication among scientists. By the early 1920s, the old quantum theory could no longer reconcile a growing set of experimental findings with classical mechanics. Heisenberg's matrix mechanics in 1925, followed by the work of Born and Jordan, replaced classical trajectories with a noncommutative mathematical formalism. Schrödinger's wave mechanics, developed independently in 1926 from de Broglie's matter-wave hypothesis, offered a formally equivalent route. Contributions by Bohr, Pauli, and Dirac helped consolidate the new mechanics and its probabilistic interpretation. These advances emerged from closely connected circles centred on Copenhagen and Göttingen, where seminars, visits, and correspondence moved ideas across institutions with unusual speed (van der Waerden 1967; Mehra and Rechenberg 1982; Kojevnikov 2020). In Kuhn's terminology, quantum mechanics is a canonical scientific revolution: it reorganized the accepted problems, standards, and conceptual foundations of physics rather than simply adding another result to the existing framework (Kuhn 1970).

The historical setting provides a common scientific shock with heterogeneous prior access. The publication of the new mechanics affected the physics profession as a whole, so the analysis cannot identify the total effect of quantum mechanics by comparing physics before and after 1925. Instead, it asks whether scientists who already corresponded with the founders experienced a larger change in output after the breakthrough than scientists without a direct link. The estimand is therefore the relative post-1925 advantage associated with pre-existing direct access to the founders. It can capture faster transmission of ideas, tacit knowledge, feedback, or entry into productive research networks. It does not capture the aggregate effect of the revolution, since unconnected scientists could learn from journals, conferences, and indirect contacts.

I construct the communication network from the *Inventory of Sources for History of Twentieth-Century Physics* (ISHTCP), an archival inventory covering approximately 5,600 physicists and nearly 750,000 letters written between 1896 and 1952 (Wheaton 1993). I digitize the physicists' identities and characteristics together with the recorded direction, volume, and period of their correspondence. Letters were an important medium for discussing scientific problems and

circulating work before publication, which makes the inventory an unusually direct record of interpersonal access. I link these data to articles and forward citations in leading physics journals indexed by the Web of Science. The resulting estimation sample is a balanced panel of 1,343 scientists observed annually from 1919 through 1932, for a total of 18,802 scientist-year observations. The eight founders are excluded from the estimation sample.

The baseline treatment indicator equals one if a scientist had a recorded correspondence link, in either direction, to at least one of eight founders before the breakthrough: Niels Bohr, Max Born, Louis de Broglie, Paul Dirac, Werner Heisenberg, Pascual Jordan, Wolfgang Pauli, or Erwin Schrödinger. This definition treats the presence of a direct link as a measure of access; it does not assume that every recorded letter transmitted a specific quantum-mechanical idea. I measure annual scientific output on both the extensive and intensive margins: whether the scientist published, the number of publications, and citations received by that year's publications over five- and twenty-five-year forward windows. Publications measure the production of research, while citation-weighted publications measure its subsequent reception and influence ([Waldinger 2012](#)).

The empirical analysis begins with a two-way fixed-effects difference-in-differences specification comparing changes after 1925 for connected and unconnected scientists. Scientist fixed effects absorb permanent differences in ability and career-long productivity, while publication-year fixed effects absorb shocks common to the sampled profession. Because connected scientists were already more productive before 1925, the preferred specifications use pre-treatment observables to make the comparison group more similar to the connected group. I estimate inverse-probability-weighted and doubly robust event-study difference-in-differences models following [Callaway and Sant'Anna \(2021\)](#) and [Sant'Anna and Zhao \(2020\)](#). The propensity score conditions on pre-1925 publication activity, fractional authorship, short-horizon citations, scientific specialty, principal country of work, birth year, and first publication year. The event-study coefficients trace the differential response by year and provide evidence on pre-treatment dynamics. Inference uses wild-bootstrap confidence intervals clustered by scientist.

The fixed-effects benchmark shows a positive post-1925 differential for directly connected scientists. Their annual probability of publishing increased by 5.4 percentage points relative to the contemporaneous change among unconnected scientists. Against a pre-period mean of 0.428 for the connected group, this corresponds to a 12.6 percent increase. The estimate for $\log(1 + \text{publications})$ is positive but imprecise. The citation estimates are larger: connected scientists' post-1925 work received 0.123 log points more citations within five years and 0.131 log points more citations within twenty-five years. The latter coefficient corresponds to an increase of approximately 14 percent in one plus citations. The evidence therefore points more clearly to a change in the production and recognition of influential work than to a precisely estimated increase in the number of papers.

The weighted event studies reinforce this interpretation while showing the uncertainty

around individual horizons. For twenty-five-year citations, the inverse-probability-weighted estimates are centred near zero before 1925 and positive in every displayed post-treatment year. The doubly robust specification produces a similar profile. Confidence intervals remain wide at several horizons, so the figures provide stronger evidence of a sustained post-1925 differential than of its exact year-by-year path. Moreover, citations to papers published before 1925 may themselves arrive after the breakthrough. Leads in a long-horizon citation outcome are therefore not clean placebo tests for anticipation; publication outcomes provide the more informative evidence on pre-treatment output trends.

Two field-specific exercises help assess whether the pattern is connected to the intellectual content of the shock. Restricting the sample to theoretical quantum studies yields a pooled estimate for twenty-five-year citations of approximately 0.13 log points, close to the full-sample benchmark, and the post-1925 event-study estimates remain positive. In contrast, a negative-control comparison using astrophysics and physical chemistry shows no sustained positive post-treatment path. This contrast is consistent with pre-existing access mattering most for scientists working near the new theory. It is nevertheless suggestive rather than a formal test of equality across fields and cannot by itself eliminate field-specific confounding.

The research design has important limitations. Correspondence links were not randomly assigned, and weighting balances only observed characteristics. Selection on changing, unobserved determinants of productivity may remain. The treatment measure records the existence of a link rather than its content or strength, and part of the archival correspondence is grouped into a 1922–1925 interval that cannot always distinguish contact formed before 1925 from contact initiated during the breakthrough year. The inventory may omit interactions through unrecorded letters, face-to-face meetings, or other channels. Finally, knowledge can diffuse through indirect network paths, so an ostensibly unconnected scientist may still be exposed through connected colleagues. Such interference can blur the difference between direct and indirect effects ([Vazquez-Bare 2023](#)). For these reasons, I interpret the results as conditional evidence of a differential change associated with direct founder access, rather than as a design-based causal estimate free of selection and spillovers.

The paper contributes to three related literatures. First, it contributes to the economics of science by studying how interpersonal access is associated with scientific output during a period of rapid intellectual change. Existing work demonstrates that peers and mentors can shape researchers' productivity and creative direction ([Waldinger 2012](#); [Borowiecki 2022](#)). The correspondence data add a measure of direct communication that does not require coauthorship, common employment, or geographic proximity.

Second, the paper contributes to research on social networks and knowledge diffusion. Network position can affect when agents receive and adopt information ([Banerjee et al. 2013](#)), while sustained interpersonal exchange can support the production and recombination of useful knowledge ([Mokyr 2016](#)). The historical correspondence network makes it possible to observe

links among scientists before a major shock and then follow their research output over time. It also exposes a central empirical difficulty in network settings: information may reach the comparison group through indirect paths, making spillovers part of the substantive mechanism as well as a threat to conventional identification.

Third, the paper provides a micro-level quantitative account of a scientific revolution. Historical scholarship emphasizes that discoveries are collective processes whose meaning and consequences develop across researchers and institutions rather than at a single instant (Kuhn 1962). By combining this historiography with scientist-level communication and output data, the paper examines how an intellectual transformation propagated through an existing professional network. This approach connects quantitative economic history to the study of innovation by treating a scientific breakthrough not only as a new stock of codified knowledge, but also as a change whose productive use may depend on prior social access.

The remainder of the paper proceeds as follows. Section 2 describes the development of quantum mechanics and the Copenhagen–Göttingen network. Section 3 introduces the correspondence, publication, and citation data. Section 4 documents the structure of the pre-1925 correspondence network. Section 5 presents the treatment definition, estimands, and identifying assumptions. Section 6 reports the fixed-effects and weighted event-study results.

2. Historical Background

The emergence and rapid development of quantum mechanics between 1925 and 1927 was the outcome of a crisis in the old quantum theory and of intense exchange among several research centres, including Copenhagen, Göttingen, Munich and Berlin. Journals, seminars, visits, proof sheets, and private correspondence circulated unfinished arguments quickly. The resulting theory changed both the mathematical language of physics and its account of what could be predicted about nature. This combination of rupture and networked discovery makes the quantum revolution a useful setting in which to study social ties and the diffusion of knowledge (van der Waerden 1967; Kojevnikov 2022).

2.1. From classical physics to the crisis of the old quantum theory

At the beginning of the twentieth century, mechanics and electrodynamics were built around continuous motion, definite trajectories, and deterministic laws. Planck’s quantum hypothesis of 1900 and Einstein’s light quantum of 1905 challenged this framework but did not supply a general mechanics of atomic processes. Bohr’s 1913 model explained the stability and spectrum of hydrogen by combining classical electron orbits with discontinuous transitions between stationary states. His correspondence principle connected the new rules to classical physics in the appropriate limit and guided much subsequent work (Heilbron and Kuhn 1969).

The Bohr–Sommerfeld theory produced striking successes, yet its mixture of classical dynamics and quantum conditions was not self-contained. By the early 1920s it could not satisfactorily explain helium, many-electron atoms, the anomalous Zeeman effect, or spectral-line intensities. Physicists accumulated increasingly elaborate rules without a consistent account of atomic motion. This crisis called for a reformulation rather than another correction of classical mechanics (Mehra and Rechenberg 1982; van der Waerden 1967).

2.2. The Copenhagen–Göttingen circle and matrix mechanics

The breakthrough grew out of a network rather than a single laboratory. Bohr’s Institute for Theoretical Physics, opened in Copenhagen in 1921, became an international meeting point for young theorists. Bohr shaped the agenda through his atomic theory, seminars, hospitality, and extensive correspondence. Göttingen supplied a complementary centre under Max Born, linked to its mathematical community and James Franck’s experimental institute. Researchers moved among Copenhagen, Göttingen, Munich, Hamburg, Cambridge, and other centres. The “Copenhagen–Göttingen school” is therefore best understood as a connected circle, not a group united on every question (Kojevnikov 2020, 2022).

Werner Heisenberg occupied a central position in this circle. Trained by Arnold Sommerfeld in Munich, he worked with Born in Göttingen and spent the winter of 1924–1925 with Bohr and Hendrik Kramers in Copenhagen. His 1925 paper abandoned hypothetical electron orbits and built a mechanics from observable transition frequencies and amplitudes. Born recognized Heisenberg’s noncommutative multiplication rule as matrix multiplication; with Pascual Jordan he formulated the canonical commutation relation, after which Born, Heisenberg, and Jordan extended the theory in their *Dreimännerarbeit*. Matrix mechanics replaced visualized trajectories with algebraic relations among infinite arrays, changing what counted as a legitimate physical description (van der Waerden 1967; Mehra and Rechenberg 1982).

The achievement remained collective. Jordan supplied essential mathematical technique. Pauli, a sharp critic whose letters connected the main centres, showed that matrix mechanics reproduced the hydrogen spectrum. Paul A. M. Dirac, working in Cambridge outside the immediate Göttingen group, independently related quantum commutators to classical Poisson brackets. The transformation theories of Dirac and Jordan then moved quantum mechanics toward its standard abstract operator language. The 1925 breakthrough was thus a rapid chain of linked advances, not a finished theory appearing in one paper (van der Waerden 1967).

2.3. Wave mechanics, duality, and the probabilistic interpretation

A second route began with Louis de Broglie, who in 1923–1924 extended the wave–particle duality of radiation to matter. Erwin Schrödinger developed this insight into wave mechanics in early 1926. His differential equation reproduced the hydrogen spectrum using tools more

familiar than Heisenberg's matrices. Schrödinger and others soon demonstrated the close formal equivalence of wave and matrix mechanics, although their physical pictures differed. Their joint success established wave-particle duality as central to the new physics: quantum objects could be described as neither classical particles nor classical waves alone ([van der Waerden 1967](#); [Bacciagaluppi and Valentini 2009](#)).

The common formalism did not settle its meaning. Schrödinger sought a continuous, realist account of the wave, while the Copenhagen-Göttingen circle increasingly interpreted the theory statistically. In 1926 Born argued that the wave function determines probabilities rather than a definite trajectory. In 1927 Heisenberg formulated the uncertainty relation: conjugate quantities such as position and momentum cannot simultaneously be arbitrarily precise. Bohr placed this limit within complementarity, according to which mutually exclusive experimental arrangements reveal different but jointly necessary aspects of a quantum phenomenon.

These ideas broke with classical determinism, but "acausality" requires care. Born, Heisenberg, and Jordan presented irreducible probability and the limitation of classical causality as fundamental lessons of quantum mechanics. Einstein, Schrödinger, and de Broglie resisted and pursued causal or realist alternatives. Historians therefore treat the reception of indeterminism not only as a consequence of equations, but also as a process shaped by intellectual commitments and scientific communities ([Forman 1971](#); [Kojevnikov 2022](#)).

2.4. The Fifth Solvay Congress and Kuhn's quantum revolution

The Fifth Solvay Congress, held in Brussels in October 1927 under the title *Electrons and Photons*, assembled the leading participants when the formalism was largely in place but its interpretation remained open. It confronted Born and Heisenberg's probabilistic mechanics, Schrödinger's wave mechanics, and de Broglie's pilot-wave theory, while giving lasting visibility to the Bohr-Einstein disputes over measurement and completeness. Bohr had introduced complementarity at Como shortly beforehand. The later image of Solvay as a decisive victory for a unified "Copenhagen interpretation" is too simple: de Broglie, Einstein, and Schrödinger remained unconvinced. Yet the congress clarified the competing positions and became a focal point in the subsequent consolidation of the Copenhagen approach ([Bacciagaluppi and Valentini 2009](#)).

Thomas Kuhn treated quantum mechanics as a paradigmatic scientific revolution. Anomalies in black-body radiation, specific heats, and the photoelectric effect produced a crisis that the established framework could not absorb. For Kuhn, a revolution is not cumulative: it reconstructs a field's problems, methods, standards, and objects from new fundamentals. Quantum mechanics did this by replacing classical trajectories with states and operators, deterministic prediction with probabilities, and a single visual model with complementary descriptions. Kuhn also stressed that a new paradigm is assimilated collectively and cannot be reduced to one scientist or instant. Accordingly, this paper treats 1925 as the sharp beginning of the new mechanics

and 1925–1927 as the broader period in which its formalism and dominant interpretation were constructed, contested, and diffused (Kuhn 1970).

3. Data

This paper draws on three main data sources. The first is an inventory of historical scientific correspondence (letters) among physicists over 1900–1952. The second source is the data on scientific publications and citations from leading physics journals from the Clarivate’s Web of Science database (WoS), published over 1900–1945. Finally, the third source is the OpenAlex database on cited references (sources cited in a given article) for articles published over the same years.

3.1. Data on historical scientific correspondence of physicists

A key and novel data used in this paper is the data on historical scientific correspondence – letters – among approximately 5,600 physicists over the period from 1896 to the middle 1950s. The data comes from an inventory (or a catalog), the *Inventory of Sources for the History of Quantum Physics (ISHTCP)* in Wheaton (1993). The inventory’s original purpose was to help historians of science find the location of materials (primarily, letters) of scientists active in physics in particular years. The inventory was created by a group of historians of science with Bruce R. Wheaton in charge at the end of 1980s, and resulted in a published microfilm index with data on physicists’ letters and an accompanying report Wheaton (1993).

The inventory has 12,698 pages in total on the microfilm reels. Each page presents a list of correspondents of the particular physicist (whose name is in bold font in the top of the page) and the number of *letters* and *pages* to and from the correspondents. The typical microfilm page gives information on the repositories where the physical letters can be found (archive, institute, museum, private collection, etc.). The information on letters to and from the correspondents is organized into year intervals, typically of half-dozen year length.

Most of the letters data on an inventory page are grouped into year intervals, typically 5- to 7-years long. The main year intervals are: pre-1900 (very rarely), 1901-1906, 1907-1913, 1914-1921, 1922-1925, 1926-1932, 1933-1938, 1939-1945, 1945-1952, post-1952 (very rarely). In a typical microfiche index card, one can see the number of letters and pages (if any) sent by physicist X to other physicists in each given year interval. For example, under a physicist Lev Davidovich Landau, one finds listed all the letters written either to or by Landau. Additionally, the inventory gives information about letters to and from other correspondents of the physicist X even if they are non-physicists. For example, under Albert Einstein one can find the information on letters exchanged by him with Charles Chaplin and F. D. Roosevelt. Figure 1 gives an example page from the inventory.¹

¹In the page which starts the listing of correspondence information for a physicist, the physicist’s name appears

INVENTORY OF SOURCES FOR HISTORY OF TWENTIETH-CENTURY PHYSICS -- (ISHTCP)											
P	C	Archive	Years: P > C exchg C > P			Years: P > C exchg C > P			Years: P > C exchg C > P		
Eve, A/Wilson, H											
		Slater, John Clarke US.Ph.01.b	----	46-52:	4/ 4	/	3/ 4	4 52----	/	/	+/
Eversheim, Paul				(1873-		)					
		Darmstaedter, Ludwig GW.Be.01.g	----	22-25:	1/	/	/				
Evjen, H. M.				(	-	)					
Ewald, Heinz				(1914-		)					
Ewald, Peter Paul				(1888-		)					
		Bernal, John Desmond GB.Ca.01.e GB.Ca.01.e	----	39-45:	1/ 2	/	1/ 1				
		Bethe, Hans Albrecht US.It.01.a	----	39-45:	3/ 3	/	3/ 3				
		53----: 1/ 1 /	----	32-38:	1/ 2	/	/	46-52:	1/ 1	/	1/ 1
		Born, Max GW.Be.01.e	----	32-38:	1/ 2	/	/	39-45:	4/ 12	/	/
		See: 53----: 3/ 6 /	----	32-38:	1/ 2	/	/	39-45:	4/ 12	/	/
		Bothe, Walther Wilhelm Georg GW.Be.03.c	----	32-38:	1/ 1	/	/				
		Bragg, William Henry GB.Lo.03.a GB.Lo.03.a	----	14-21:	1/ 4	/	/	22-25:	1/ 1	/	3/ 3
		32-38: 1/ 2 /	----	14-21:	1/ 4	/	/	22-25:	1/ 1	/	3/ 3
P	C	Archive	Years: P > C exchg C > P			Years: P > C exchg C > P			Years: P > C exchg C > P		
x/y = x letters on y pages; + = unknown number of letters; P > C = physicist (or C > P = correspondent) is writer, or exchanged. page 3848											
M 6											

M 6

FIGURE 1. A page from the microfilm reel in the ISHTCP

Note: The figure presents a typical page from the microfilm reel inventory (page 3,848 out of 12,600+). The colors have been made black-and-white, and noise and artifacts in the image have been removed with Python.

Source: Wheaton (1993)

To the best of my knowledge, the ISHTCP is the largest source of collected information on scientific correspondence exchange between scientists in the twentieth century. [Wheaton \(1993\)](#) notices himself that the scale of the inventory is unprecedented. The goal of the inventory was to facilitate studying the history of physics, with a particular focus on the history of quantum physics. Overall, the inventory presents information on nearly 750,000 letters written to or by physicists active in the period from 1896 to 1952. The main criterion for their selection of a physicist was that the scientist published significantly in one of the seven leading journals of pure physics in this period: *Annalen der Physik*; *Annales de Physique*; *Philosophical Magazine*; *Proceedings of the Royal Society of London A*; *Physical Review*; *Uspekhi Fizicheskikh Nauk*; and *Zeitschrift für Physik*.

The universe of physicists selected into the inventory consists of just over 5,600 physicists from all major fields in physics. Therefore, the list of physicists is a universe of the most important scientists in the twentieth century physics from all over the world. This is fine for the purposes of my paper since in science it is often true – and especially for the science in the first half of the twentieth century – that the most important innovations are created by a few individual super-star scientists, and most important and influential works are published in top journals. The revolutionary quantum mechanics is a brilliant example of this.

[Wheaton \(1993\)](#) underlines that the design of the ISHTCP is such that it reports all letters, regardless of collection, that a specified physicist exchanged with a given correspondent.

Although this paper is about the quantum physics and in particular quantum mechanics revolution, the inventory is not biased towards quantum physicists but rather covers rather all main areas in pure physics, including specialists in nuclear, solid-state, plasma, and low-temperature physics as well as the principal early contributors to acoustics, astrophysics, biophysics, and geophysics (([Wheaton 1993](#))).

A real concern with respect to the data source is that the inventory is limited by coverage of correspondence of mainly theoretical physicists, keeping the data on scientific correspondence of most technical and applied physicists excluded. However, since the main analysis of this paper is built around the quantum mechanics breakthrough, which in its early years of development was largely theoretical achievement, this selection bias should not be of much relevance for my analysis.

It is important to notice that the inventory provides information on correspondence of mainly scientific nature. In the first half of the twentieth century, even after emergence of telegraph and telephone, mail correspondence was actively used by scientists to discuss science-related matters and exchange ideas and novel results in their profession until early 1960s. Since the inventory does not provide description of topics discussed in the letters, it is hard to judge about the exact content of particular letters until investigation of the letters themselves. However,

in larger bold font on the top of the page (e.g., *Einstein, Albert*). However, since most pages are continuations, in these pages the name of the physicist for whom the correspondence information is listed in the page is written in the form *Einstein, A/Bohr, N* carrying the last correspondent's name from the previous page after the character "/".

the year intervals into which the letters were bunched, roughly reflect the main topics that had been most likely discussed in the letters in a particular interval.

3.1.1. Usage of the correspondence data

I use the data on letters exchanged between physicists to track pre-existing connections with the scientists who laid the foundations of the future quantum mechanics by publishing the first key papers on it. I call these scientists founders. To identify the founders, I consulted various works of historians of science, including articles and most monumental studies of the development of quantum physics/mechanics, such as [Mehra and Rechenberg \(2000\)](#). The data on correspondence among physicists allows me to construct various measures of connectedness based on the intensity and frequency of scientific letters exchanged. What is crucial, is that the data allows me to construct the social networks of scientists based on their inter-personal communication, by different periods in the development of physics.

The inventory is supplemented by the book-report [Wheaton \(1993\)](#). It provides a user guide to the microfiche reel index (white and yellow tapes) and consists of several sections, including general statistics on social correspondence networks. Importantly, it has a section with a list of physicists' names that appear in the index. Figure 2 presents two typical pages from the *Physicists* section in the book with the physicists' names that appear in the main microfiche index. The inventory also contains a separate yellow-color card index which separately lists all non-physicists' correspondents for each physicist name in the main index.

I digitize the full list of physicists' names, together with their attributes (life years, country of work, country of origin, and specialization), from the section "Physicists" (120 pages, 5,600+ names) and construct a dataset of the names. This is the main dataset which represents the universe of physicists that I study. Later, however, the universe shrinks to a lesser number of names because the period of my analysis is from 1919 to 1932, and not all of the physicists listed in the book-report were actively and alive in the period.

3.1.2. Collecting and digitizing of the correspondence data

The original correspondence data was accessed via an inventory and a report on microfilm reels in the Technical University of Dresden Library, freely accessible for public. I used the special scanning machine which was able to read the microfilm reels, zoom in if necessary, and made snapshots in PDF save immediately on a USB-folder. I processed exactly like this, spending approximately one month to scan the content of the full set of microfilm reels.

Inventory of Sources for History of Twentieth-Century Physics					Physicists				
Coggeshall, Norman David	(1916-)	TT	US	21	Cool, Rodney Lee	(1920-)	NU	US	18
Cohen, Bernard Leonard	(1924-)*	NU	US	21	Cooley, Leroy Clark	(1833-1916)*	TT	US	19
Cohen, E. Richard	(-)*	NU	US	32	Coolidge, Albert Sprague	(1894-)*	PC	US	17
Cohen, Ernst Julius	(1869-1944)*	PC	Ne	10a	Coolidge, William David	(1873-1975)*	TT	US	19
Cohen, Louis	(1876-1948)*	TT	US	22	Coomes, Edward Arthur	(1909-)*	SS	US	26
Cohen, Victor W.	(1911-)*	NU	US	19	Coon, James Huntington	(1914-)	NU	US	31
Cohn, Emil Georg	(1854-1944)*	TT	Ge	4a,11a	Cooper, Daniel	(-)	NU	US	17
Cohn, Willi M.	(-)*	QG	Ge US	3b,33	Cooper, Eugene Perry	(-)*			
Cohn-Peters, Hans Juergen	(1905-)*	TT	GB		Cooper, Franklin Seany	(-)*			
Coker, E. G.	(-)*	SS	US	31	Cooper, John N.	(-)*	QG	US	25
Colby, Malcolm Y.	(1880-1970)*	TT	US	23	Cope, Thomas Darlington	(1880-)*	TT	US	21
Colby, Walter Francis	(1861-1928)*	TT	US	25	Copeland, Paul Lavern	(1905-1964)*	QG	US	27
Cole, Alfred Dodge	(-)*	TT	US	16	Copson, Edward Thomas	(-)*	TT	GB	
Coleman, Howard S.	(1917-)	OP	US	31	Corben, Herbert Charles	(1914-)*	QG	Ca	15
Coleman, Richard L.	(-)	QG	US	35	Corbino, Orso Mario	(1876-1937)*	QG	It	8a
Coles, Donald Kennedy	(1911-)	TT	US	26	Cordes, Heinrich	(1906-)	TT	Ge	4b
Colgate, Stirling Auchincloss	(1925-)*	NU	US	31	Cork, Bruce	(1915-)	NU	US	27
Collie, C. H.	(-)*	NU	GB		Cork, James M.	(1894-1957)*	QG	US	28
Collie, J. Norman	(1859-1942)*	TT	GB		Cornell, Samuel D.	(-)*	PC	US	23
Collins, Ernest Hobart	(-)*	QG	US	30	Cornier, J.	(-)*	QG	GB	
Collins, George Briggs	(1906-)*	NU	US	24	Cornog, Isaac	(-)	QG	US	21
Collins, Joseph R.	(-)	PC	US	23	Correll, Malcolm	(1914-)	NU	US	30
Coltman, John W.	(-)*	NU	US	21	Corson, Dale Raymond	(-)*	NU	US	19
Colwell, Robert Cameron	(1883-1959)*	QG	US	24	Corson, Edward M.	(1921-)*	QG	US	22
Compton, Alfred George	(1835-1913)*	TT	US	18	Costanzo, Giovanni	(1874-)*	QG	Pt	8a
Compton, Arthur Holly	(1892-1962)*	NU	US	27	Coster, Dirk	(1889-1950)*	QG	Ne	10a
Compton, Karl Taylor	(1887-1954)*	QG SS	US	17,20	Cotton, Aime Auguste	(1869-1951)*	TT	Fr	11a
Comstock, Daniel Frost	(1883-1970)*	TT	US	17	Cotton, Emile Clement	(1872-1950)*	NU	Fr	11b
Conant, James Bryant	(1893-1978)*	PC	US		Coudres, Theodor Des	(1862-1926)*	TT	Ge	6a
Condon, Edward Uhler	(1902-1974)*	QG	US	30	Coulson, Charles Alfred	(1910-1974)*	PC	GB	1a
Conn, G. K. T.	(-)	TT	GB		Coulson, John	(-)*	OP	US	24
Conn, William M.	(1897-)	TT	US	29	Courant, Ernest David	(1920-)*	NU	US	19
Conrad, Victor	(1876-1962)*	AP	Au US	7a	Courant, Richard G.	(1888-1972)*	MA	US	18
Constable, F. H.	(-)*	PC	GB		Courvoisier, Leo	(1873-)*	AP	Fr Ge	3a,36
Constant, Frank Woodbridge	(1904-)*	TT	US	16	Couture, L.	(-)	SS	Fr	11b
Convers, Marcello	(1917-1988)	NU	It	8a	Cowan, Clyde Lorrain, Jr.	(1919-)*	NU	US	22,29
Conway, Arthur William	(1875-1950)*	TT	Ir	2a	Cowan, Eugene Woodville	(1920-)	TT	US	34
Coode-Adams, W. R. C.	(-)	OP	GB		Cowie, Dean Bruce	(-)*	NU	US	22
Cook, Clarence Sharp	(1918-)*	QG	US	31	Cox, Ernest Gordon	(-)*	SS	GB	
Cook, James M.	(1894-1957)*	QG	US	35	Cox, Everett Franklin	(1908-)	NU	US	24
Cook, Richard Kaufman	(1910-)*	TT	US	23	Cox, Henry L.	(-)*	SS	GB	
Cook, Samuel Richard	(-)*	QG	US	32	Cox, Richard Threlkeld	(1898-)*	TT	US	22
Cooksey, Charlton Downs	(-)*	TT	US	16	Coxeter, Harold Scott Macdonald	(1907-)*	MA	Ca	15
Cooksey, Donald	(-)*	NU	US	33	Craig, D. P.	(-)	PC	GB	

FIGURE 2. Two pages from the *Physicists* section in the ISHTCP report.

Note: The figure presents two pages, 40-41. The columns in the appearing order are: physicist name, birth and death years, primary specialty(ies) (field) of work in physics, primary country(ies) of work, microfiche reel number indicator. The asterisk after the lifetime years indicates that the index contains info on letters of the physicist.

Source: Wheaton (1993)

TABLE 1. Earliest publication years available in the Web of Science database by journal

Journal	Earliest year in WoS
<i>Annalen der Physik</i>	1900
<i>Annales de Physique</i>	1946
<i>Philosophical Magazine</i>	1900
<i>Proceedings of the Royal Society of London Series A</i>	1905
<i>Physical Review</i>	1900
<i>Uspekhi Fizicheskikh Nauk</i>	1955
<i>Zeitschrift für Physik</i>	1920
<i>Physikalische Zeitschrift</i>	1900
<i>Die Naturwissenschaften</i>	1913
<i>Nature</i>	1900
<i>Science</i>	1900

Notes: The table reports the earliest publication year available in Clarivate's Web of Science (WoS) for each journal. No records for *Uspekhi Fizicheskikh Nauk* are available in WoS before 1955.

3.2. Publications and citations data

I download data on scientific publications of document type “article” published in the period 1919–1932, and data on their (forward) citations up until 1952 from the Clarivate's WoS database.² I start in 1919, first year post-WWI, to exclude articles that were likely affected by the war and changes in international scientific collaboration.³ I end in 1932 to eliminate possible bias related to the expulsion of Jews-origin and "politically unreliable" scientists by the Nazi government starting in 1933.

I select a set of academic journals which includes leading academic journals in physics, and general interest journals of that time. The journals are: *Annalen der Physik*; *Annales de Physique*; *Philosophical Magazine*; *Proceedings of the Royal Society of London A*; *Physical Review*; *Uspekhi Fizicheskikh Nauk*; *Zeitschrift für Physik*; *Physikalische Zeitschrift*; *Die Naturwissenschaften*; *Nature*; and *Science*. These journals cover not only German-speaking, but English speaking countries. The articles from these journals cover all major fields of pure physics. I download data on publications and forward citations from the sources in the Web of Science. Table 1 shows the coverage of the selected journals in WoS, i.e. the earliest year of publication.

I realize that the focusing on only leading pure journals in physics is limiting my analysis of the scientists' general productivity.

One of the research questions in this paper is what is the effect of being in a social proximity to the superstar, key founders of a break-through new theory in science – the quantum mechanics – on subsequent general productivity of scientists? To answer this question, I introduce several

²I used Charles University's subscription to access the data.

³Iaria, Schwartz and Waldinger (2020, JPE) show that the WWI and the following boycott of German science led to significant disruption in the production of scientific knowledge and cross-citations between scientists in the war camps.

measures of scientist-level output productivity. The first is simply the number of publications (of type “article”) published by a scientist in a given year. The second is the publication share of a scientist – it counts the number of publications per scientist in a given year adjusted by the weight calculated as the 1 over the number of authors per paper. For instance, if a published article has three authors, each of them receives the publication share 1/3 for this article, irrelevant of whose name is the first in the authors list. These variables reflect only the quantity side of the productivity of a researcher.

The third measure is the citations – or forward citations – for a set of scientist’s publications published in a given year, calculated for different time windows (5, 25 and 50 years since $\tau + 1$ where τ is the publication year cohort). The citations measure is calculated as the the total number of citations coming from all sources (meaning not only physics but all other disciplines) in the world in a given year. This is a baseline measure of scientific productivity that reflects quality of the article (recognition by the scientific community in a very broad sense). [Waldinger \(2012\)](#) calls this measure *citation weighted publications*, though it is simply the count of total citations per article in a given year coming from all sources in the world. For my baseline results about the general scientist productivity, I use the measures of scientist output closely similar to approach by [Waldinger \(2012\)](#), though I explore alternative definitions in the rest of the paper.

Aghion et al. (2024) and Pauskiewicz and ... (2026, JMP) use the same or similar measures of scientific output.

I merge the physicist dataset with the author-level publications and citations dataset from the WoS. I match the names from the physicists dataset with the authors from the WoS dataset by using a combination of fuzzy matching techniques in Python. For details, see ???. As a result, I obtain a balanced panel dataset of scientists included in the list of the ISHTCP index and their individual-level data on characteristics and scientometrics outcomes, for the period 1919-1932.

3.3. Cited references data

To study questions about the diffusion of knowledge, I need data not only on forward citations (total citations that an article received in a given year from all source) but on the cited references – the list of sources (other articles) cited in a given article. The cited references data will allow to track cross-citations between the papers and authors - who is citing whom, and which year. Obviously, the mutual citations are not a perfect measure of the diffusion of knowledge, and alternative measured based on the titles and abstracts/texts of papers could be better ones for studying the diffusion of particular concepts and ideas. However, WoS provides full abstract, not even mentioning text of paper, for a very few articles in the period of my study; and even the OpenAlex database, although giving abstracts for more papers than WoS, also has lots of missing data. Therefore, for simplicity and for preliminary analysis, I rely on the data on cited references.

Extracting data on cited references is not a straightforward task to do in the WoS web UI.

The use of API provided to the subscribed universities also does not help.⁴ Instead, I rely on free open-source database of scientific works and citations – OpenAlex.⁵ I access the database and download all data on works of type “article” and their cited references, as well as authorship, affiliations of authors and topics and concepts (when available) for the works published in top journals in physics in the period 1900–1952. The journals are: **place list of journals**.

The data on cited references is invaluable for studying the diffusion of knowledge in science because it provides information on who cites whose papers, which can be considered as a measure of reception of/ access to the codified knowledge carried by the cited paper.

3.4. Other data on physicists

For additional data on physicists – biographical data and daily correspondence and info on unpublished manuscripts – I rely on another catalog of scientific correspondence and manuscripts – *Sources for the History of Quantum Physics (SHQP)* in the Kuhn et al. (1967). The full content of the book can be accessed in the digital form on the website of the American Philosophical Society.⁶ The microfilm catalog contains a collection of materials – a selected list of manuscripts and – primarily – letters of 280 physicists who contributed at least marginally to quantum studies in physics. The catalog is incomplete and misses information on some letters. However, the advantage of the data is that the data is presented by dates of sent/received letters, and that for each physicist, there is a schematic biography with all career movements information available at the end date of the SHQP project.

The catalog was collected by T. Kuhn, P. Forman, J. Heilbron and L. Allen in the project under the original name *Sources for history of quantum physics: An inventory and report* (Philadelphia, 1967) published in the form of a book. This was a pioneering project of the scale which goal was to preserve the information about the storage of unpublished materials related to the history of quantum physics across libraries, universities, archives and private collections all over the world. The resulting catalog is an index and a report to the inventory of sources for studying the history of quantum physics. The largest and primary part of the catalog is scientific correspondence (letters).

The AHQP project focused on preserving information on those manuscripts and letters which had not been published by the time of the project, and the authors of the documents had to be physicists whose work at least occasionally concerned quantum physics. As a result, the project covered 280 names, of which all most important contributors to quantum physics, such as M. Planck, W. Heisenberg, E. Schrödinger, L. de Broglie, P. Jordan, P. Dirac, W. Pauli, N. Bohr, A. Einstein and others are necessarily included. The list of names was created based on

⁴To the best of my knowledge, the WoS API allows to extract references for maximum one document ID at once, which makes it impractical to collect cited references for thousands of articles.

⁵I accessed the SQL slice of the OpenAlex database on the heavyhorse stations at CERGE-EI.

⁶<https://www.amphilsoc.org/guides/ahqp/>. One can also meet the alternative name of the source – AHQP, namely the Archive for the History of Quantum Physics.

the condition that a physicist included must have contributed at least occasionally to the topics on quantum studies.

The most covered period of correspondence is approximately in 1900-1930s. Most of time, the catalog gives information on scientific correspondence of physicists in the following format:

- The number of letters sent from physicist X to physicist Y , together with the exact calendar date of each letter.
- The archival source of the correspondence, i.e., the collection in which the original letters are physically preserved.

A typical entry in the Authors section is a name of a physicist under whose name there are surnames of other physicists also presented in the catalog, and the dates of letters sent by the author to the physicists. An individual entry (date) is an individual letter. See Figure A.4 in the Appendix for a typical author entry on the website. The AHQP data sets of scientific correspondence of 280 quantum physicists has few important advantages as well as some drawbacks. The advantages are that the dataset gives detailed information on letters exchanged between physicists on a daily basis. The level of detail allows us to build networks in dynamics.

The main drawback of the AHQP catalog data is that the data is only a part of full scientific correspondence presented in depositories all over the world. To remind, the goal of the project was to gain information on manuscripts which have not been cataloged properly by the start of the project in 1962. Hence, although the social graphs built on this data can provide interesting descriptive statistics, there are likely a significant number of missing connections as well as the total number of letters exchanged between physicists can be missing. The drawback of the AHQP data can be down-scaled to a large degree by exploring only a part of the data, more precisely the data in which `**continue**`

It is important to make a note on the relation of the ISHTCP inventory and the AHQP: the latter is entirely subsumed in the ISHTCP inventory ([Wheaton 1993](#)).

I have web-scraped the full data from the AHQP website with the help of Python, and combined it into properly structured dataframes. I extract schematic biographies of the 280 physicists provided in the AHQP.

4. Correspondence Social Networks (1900-1925)

This section provides a brief descriptive analysis of the correspondence social networks for a subset of incumbent physicists in the period 1900-1925.

4.1. Construction of the incumbent scientists sample and their correspondence links

I construct the incumbent set of physicists and their correspondents (only physicists) as follows. First, I restrict my dataset of edges to keep only edges for the years pre-1925, i.e. the

years 1900-06, 1907-13, 1914-21, and 1922-25. Including correspondence of 1922-25 is somewhat poses a risk of including scientists who selected themselves already after the arrival of quantum mechanics. I perform a robustness exercise without the links formed in 1922-25 in the Appendix.

Before dealing with links in the network, I restrict my set of physicists to be scientists who published their first paper between 1900 and 1924, and they stay alive in 1925, the year of discovery of matrix mechanics (Heisenberg 1925).⁷ After applying these filters, I obtain the total number of incumbent physicists, up to the fifth neighbors, to be 1,626.

My incumbent physicists sample is built starting from the core – the "fathers of quantum mechanics": Niels Bohr, Werner Heisenberg, Max Born, Pascual Jordan, Wolfgang Pauli, Paul Dirac, Erwin Schrödinger and Louise de Broglie; and Albert Einstein. These scientists contributed outstandingly to quantum studies. Niels Bohr and Albert Einstein are always in the top-5 physicists by their total degree, betweenness and eigenvector centrality in the pre-1925 network. Starting from the core, I track all their neighbors up to the fifth order of the neighborhood: the first, the second, the third, the fourth and the fifth neighbors of the core. The first, second, third, fourth and fifth sets of neighbors constructed in a way that they are not overlapping sets of physicists. These core and the neighbors comprise *the giant connected component* of Niels Bohr, the other founding fathers of quantum mechanics, and Albert Einstein.

It is important to notice that all other important physicists, such as Arnold Sommerfeld, Max Planck, Max von Laue, etc. automatically fall into the giant component: they are mostly appearing as the second neighbor edges.

The Figure 4 presents the graph of the full pre-1925 network of physicists: the core, the first, the second and the third order neighbors. I also picture the graphs for periods 00-06, 07-13, and 14-21 separately in the Figures in the Appendix.

The Table 2 gives counts of unique physicists by correspondence-network set. One can see that the network is expanding very fast – the largest set is the second neighbors set, – but it is rather shallow too: in the fifth neighbor set, we have only single physicist. This means that on the fifth order of neighborhood, the giant component of Bohr and Einstein becomes practically exhaustive.

In network terms, these numbers suggest a rapidly expanding but relatively shallow correspondence network around the core physicists. The first-neighbor group consists of 344 scientists directly connected to the core, while the second-neighbor group is substantially larger, comprising 1,085 scientists. In contrast, the number of more distant scientists declines sharply to 165 third neighbors, 8 fourth neighbors, and a single fifth neighbor. Thus, most scientists in the connected component are within two steps of the core, indicating a relatively compact network in terms of path length. The sharp decline beyond the second-neighbor group further suggests network saturation, as most reachable scientists have already been identified within two degrees of separation. While these descriptive statistics alone are insufficient to establish

⁷The first year of publication *fy* is extracted from articles only from a set of the seven leading journals in physics over 1900-1945, i.e. one can treat the *fy* as the year when one publishes in one of the elite physics journals.

TABLE 2. Number of unique physicists by correspondence-network group, 1900-1925

Network group	Number of unique physicists
First neighbors	344
Second neighbors	1,085
Third neighbors	165
Fourth neighbors	8
Fifth neighbors	1
Core physicists	9
Total physicists	1,612

Notes: The core physicists are Bohr, Born, de Broglie, Dirac, Einstein, Heisenberg, Jordan, Pauli, and Schrödinger. Neighbor groups are defined according to the shortest-path distance from the core physicists in the historical correspondence network. The neighbor sets are non-overlapping with each other.

that the correspondence network exhibits a small-world structure, they are consistent with a relatively shallow connected network.

Table 4 presents network coverage from Bohr and Einstein across correspondence periods. Table 6 reports the top-20 Bohr’s strongest direct correspondence connections, 1900–1925. Similarly, Table 7 reports the same information for Einstein.

Figure A9 in the Appendix presents the same network as in Figure 5 but now visualized using the ForceAtlas2 layout algorithm, which is particularly suitable for larger networks and for displaying their community structure. The algorithm treats connected nodes as attracting one another, while all nodes exert a repelling force that prevents them from occupying the same position. Consequently, scientists with many connections among themselves tend to form visible clusters, whereas weakly connected groups are positioned farther apart. A gravitational force keeps the overall network together. The layout was run for 1,000 iterations, giving node positions more time to stabilize, and a fixed random seed was used to make the resulting visualization reproducible. The observed clusters should be interpreted as visual indications of network structure rather than formally estimated communities.

4.2. Star-networks of Niels Bohr and Albert Einstein

Essentially, the networks provided above are mostly the networks of two physicists – Bohr and Einstein. I therefore continue the further descriptive analysis around these two giant nodes.

Below, Figure 7 shows the social graph of first order connections of Niels Bohr and Albert Einstein which deliberately excludes connections among the neighbors themselves, so the overlap is visually clear.

The overlap between Bohr and Einstein correspondence circles can be shown as the Jaccard similarity. The Jaccard similarity between the two networks is: 0.166. Bohr had 180 direct correspondence neighbors, while Einstein had 213. They shared 56 common neighbors, representing



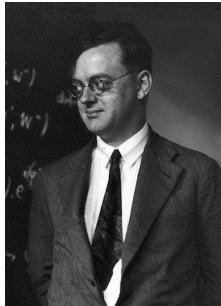
Heisenberg



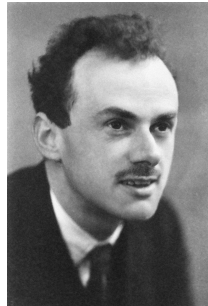
Born



Bohr



Jordan



Dirac



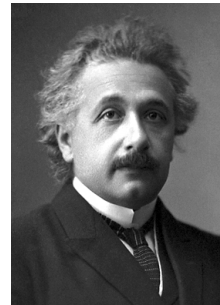
Pauli



Schrödinger



de Broglie



Einstein

FIGURE 3. Founders of quantum mechanics and Albert Einstein.

Source: Portraits obtained from Wikipedia.

TABLE 3. Top 20 physicists by alternative network-centrality measures, 1900–1925

Rank	Total degree	Betweenness centrality	Eigenvector centrality	PageRank centrality
1	Theodor L. Lyman [353] US; TT; 1874–1954	Theodor L. Lyman [0.100110] US; TT; 1874–1954	Johannes Stark [0.246392] GE; QG; 1874–1957	Johannes Stark [0.026434] GE; QG; 1874–1957
2	Johannes Stark [304] GE; QG; 1874–1957	Albert Einstein [0.074666] GE SZ US; QG; 1879–1955	Theodor L. Lyman [0.224546] US; TT; 1874–1954	Albert Einstein [0.020742] GE SZ US; QG; 1879–1955
3	Albert Einstein [293] GE SZ US; QG; 1879–1955	Niels Bohr [0.069173] DK; QG; 1885–1962	Niels Bohr [0.214892] DK; QG; 1885–1962	Theodor L. Lyman [0.018707] US; TT; 1874–1954
4	Niels Bohr [269] DK; QG; 1885–1962	Robert A. Millikan [0.063365] US; QG; 1868–1953	Robert A. Millikan [0.213846] US; QG; 1868–1953	Niels Bohr [0.016479] DK; QG; 1885–1962
5	Hendrik A. Lorentz [260] NE; QG; 1853–1928	Johannes Stark [0.054867] GE; QG; 1874–1957	Ernest Rutherford [0.209792] CA GB; NU; 1871–1937	Ernest Rutherford [0.015967] CA GB; NU; 1871–1937
6	Robert A. Millikan [250] US; QG; 1868–1953	Hendrik A. Lorentz [0.051542] NE; QG; 1853–1928	Hendrik A. Lorentz [0.203571] NE; QG; 1853–1928	Hendrik A. Lorentz [0.015071] NE; QG; 1853–1928
7	George E. Hale [243] US; AP; 1868–1938	George E. Hale [0.046475] US; AP; 1868–1938	Albert Einstein [0.197950] GE SZ US; QG; 1879–1955	Robert A. Millikan [0.013056] US; QG; 1868–1953
8	Arnold Sommerfeld [191] GE; QG; 1868–1951	Arnold Sommerfeld [0.033304] GE; QG; 1868–1951	Arnold Sommerfeld [0.184754] GE; QG; 1868–1951	Arnold Sommerfeld [0.012267] GE; QG; 1868–1951
9	Ernest Rutherford [172] CA GB; NU; 1871–1937	Otto H. Wiener [0.025947] GE; TT; 1862–1927	George E. Hale [0.176016] US; AP; 1868–1938	Joseph John Thomson [0.011175] GB; QG; 1856–1940
10	William F. Meggers [154] US; QG; 1888–1966	William F. Meggers [0.025866] US; QG; 1888–1966	Paul Ehrenfest [0.153075] AU UR NE; QG; 1880–1933	Thomas J. J. See [0.009602] US; TT; 1866–1962
11	Joseph John Thomson [148] GB; QG; 1856–1940	William Henry Bragg [0.025005] AL GB; QG; 1862–1942	Joseph John Thomson [0.139997] GB; QG; 1856–1940	Paul Ehrenfest [0.009595] AU UR NE; QG; 1880–1933
12	Paul Ehrenfest [142] AU UR NE; QG; 1880–1933	William Thomson [0.023476] GB; MA; 1824–1907	Henry Crew [0.139142] US; TT; 1859–1953	Owen W. Richardson [0.009511] GB US; QG; 1879–1959
13	Charles E. Mendenhall [137] US; QG; 1872–1935	Joseph John Thomson [0.023430] GB; QG; 1856–1940	Owen W. Richardson [0.125628] GB US; QG; 1879–1959	Joseph Larmor [0.009374] GB; QG; 1857–1942
14	William Henry Bragg [136] AL GB; QG; 1862–1942	Paul Ehrenfest [0.021285] AU UR NE; QG; 1880–1933	Heike Kamerlingh Onnes [0.122163] NE; SS; 1853–1926	George E. Hale [0.009322] US; AP; 1868–1938
15	Otto H. Wiener [135] GE; TT; 1862–1927	Owen W. Richardson [0.020893] GB US; QG; 1879–1959	Joseph Larmor [0.101858] GB; QG; 1857–1942	Svante Arrhenius [0.007997] SD; PC; 1859–1927
16	Joseph Larmor [134] GB; QG; 1857–1942	Karl T. Compton [0.020114] US; QG SS; 1887–1954	Charles E. Mendenhall [0.115711] US; QG; 1872–1935	Heike Kamerlingh Onnes [0.006715] NE; SS; 1853–1926
17	Owen W. Richardson [132] GB US; QG; 1879–1959	Joseph Larmor [0.018313] GB; QG; 1857–1942	Raymond T. Birge [0.101858] US; QG; 1887–1980	William F. Meggers [0.006638] US; QG; 1888–1966
18	Karl T. Compton [127] US; QG SS; 1887–1954	Heike Kamerlingh Onnes [0.017499] NE; SS; 1853–1926	William Henry Bragg [0.091549] AL GB; QG; 1862–1942	Raymond T. Birge [0.006617] US; QG; 1887–1980
19	Heike Kamerlingh Onnes [122] NE; SS; 1853–1926	Svante Arrhenius [0.016733] SD; PC; 1859–1927	Thomas J. J. See [0.090716] US; TT; 1866–1962	Carl Barus [0.006552] US; PC; 1856–1935
20	William Thomson [120] GB; MA; 1824–1907	Raymond T. Birge [0.015741] US; QG; 1887–1980	Carl D. T. Runge [0.090494] GE; QG; 1856–1927	Marie Skłodowska Curie [0.006423] FR; NU; 1867–1934

Note: Scientists are ranked in descending order within each centrality measure; the corresponding centrality value appears in square brackets. The second line of each entry reports country code, specialty code, and birth and death years. All measures were calculated on the directed and unweighted correspondence network for 1900–1925. Each directed correspondence tie therefore receives equal weight, and the number of letters exchanged between a pair of scientists is ignored. Total degree is the sum of in-degree and out-degree. Betweenness measures the extent to which a scientist lies on directed shortest paths; eigenvector centrality assigns higher values to scientists connected to other central scientists; and PageRank reflects the importance of incoming ties from other important scientists.

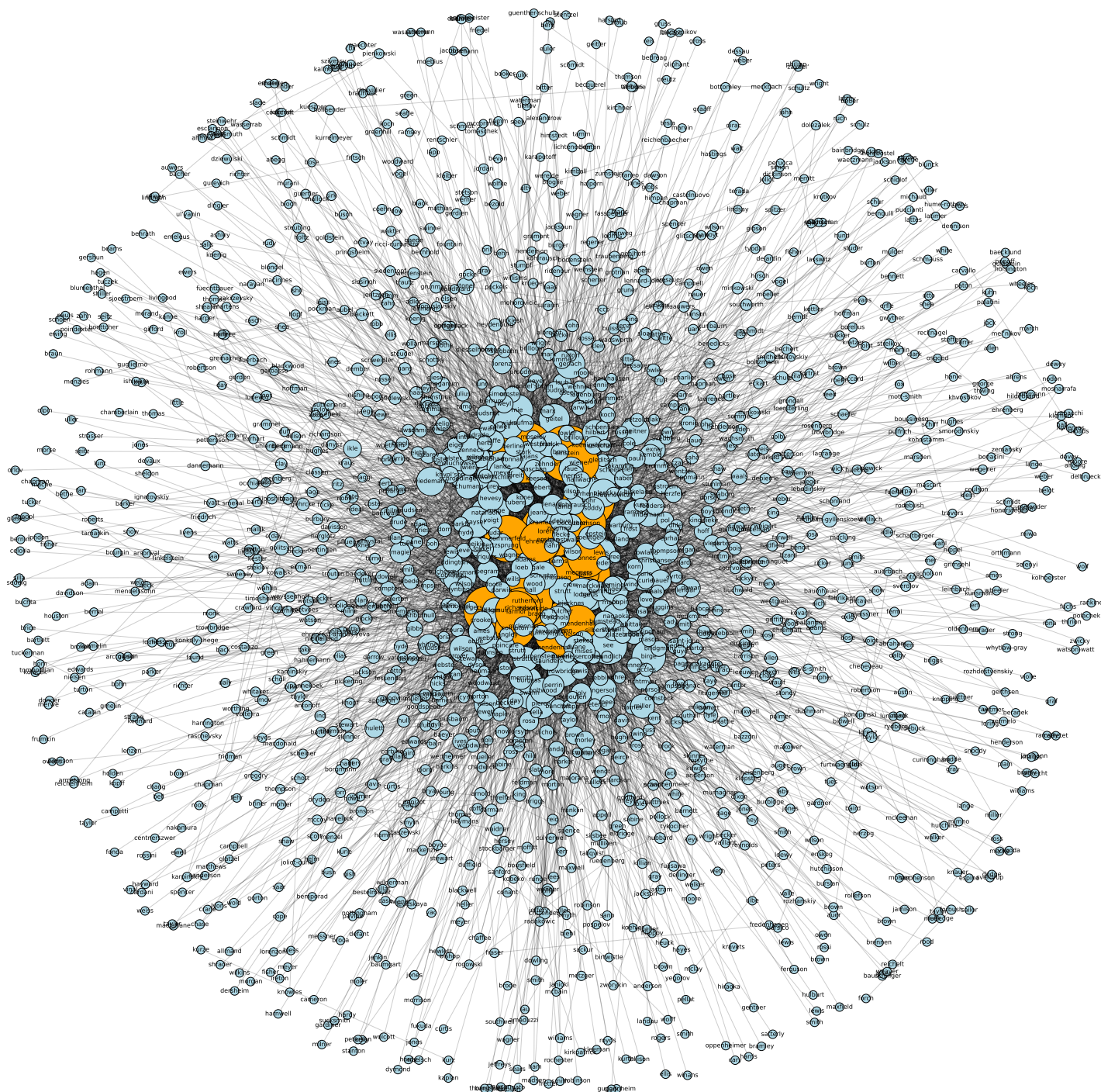


FIGURE 4. Social graph of correspondence, 1900-1925.

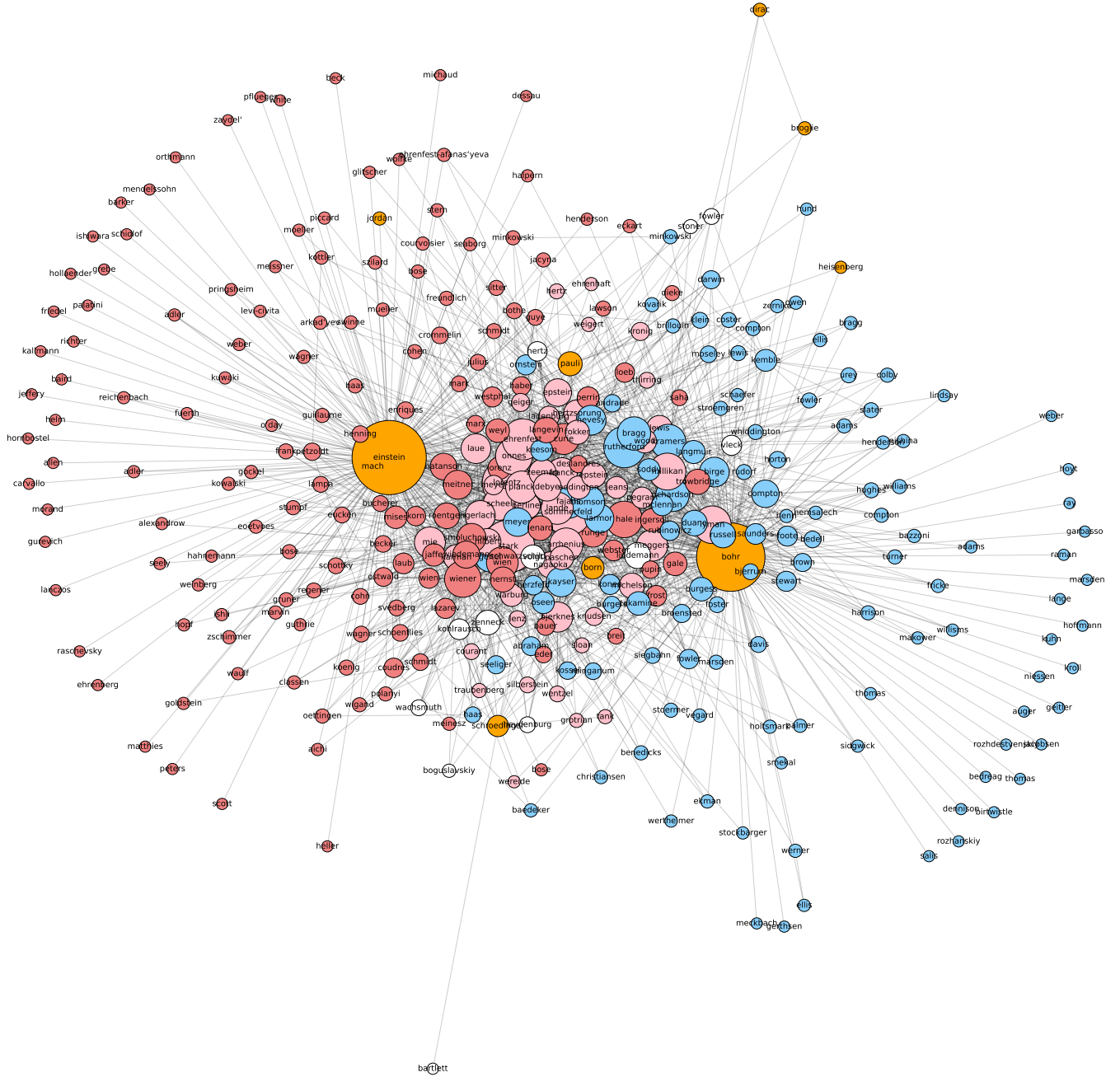


FIGURE 5. Social graph of the core and their first neighbors, 1900-1925.

Note: Undirected network. Years 1900-06, 1907-13, 1914-21, and 1922-25 are bunched together. The size of the nodes reflects the degree of a node (in or out) – the total number of connections. Colors: core (orange) vs. first neighbors. First neighbors: Bohr-only neighbors (lightskyblue), Einstein-only neighbors (coral), Bridges: Bohr- and Einstein-neighbors (pink). Spring layout algorithm, $k=0.80$, iterations=300. Graph built using Python networkx library.

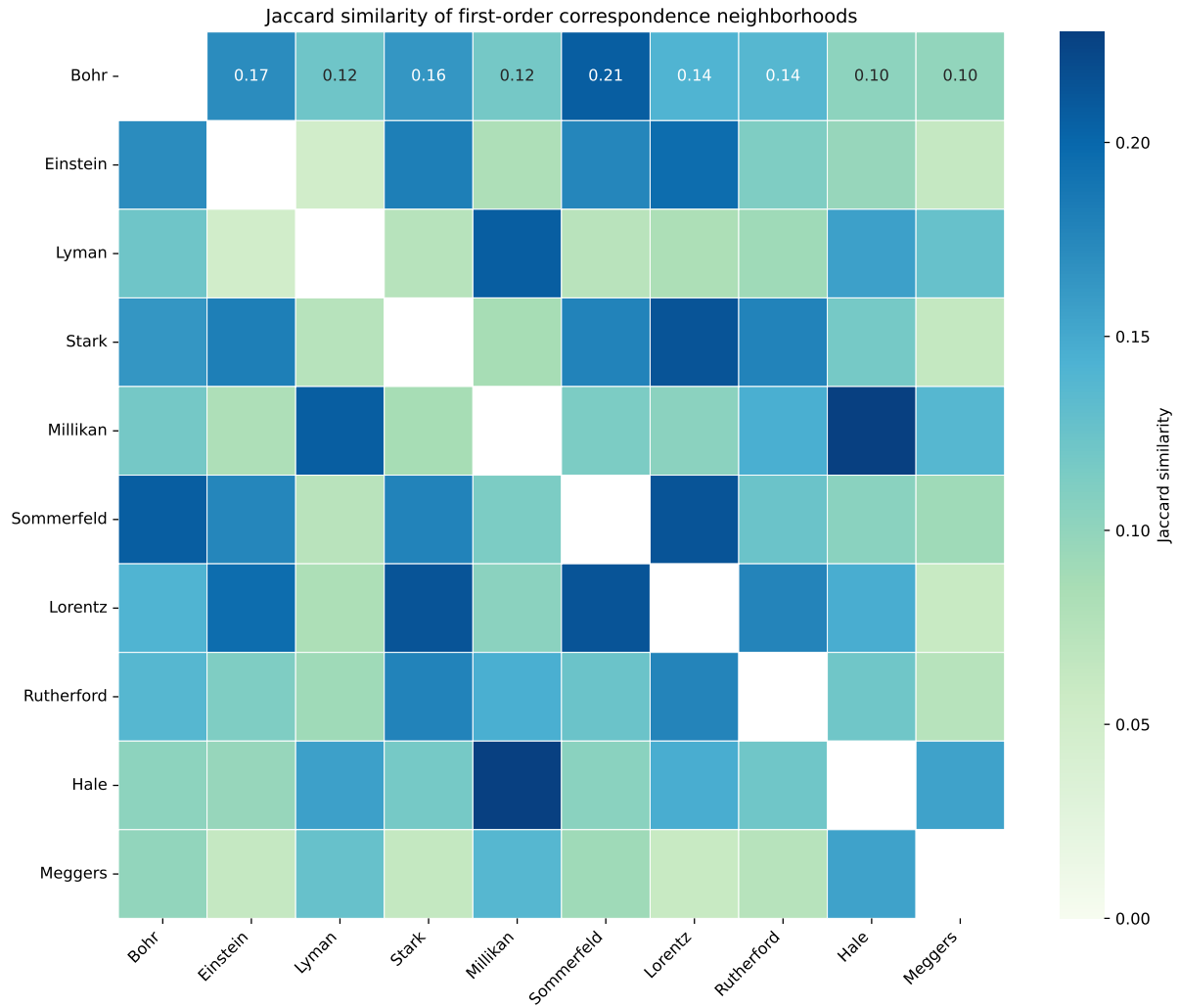


FIGURE 6. Heatmap of first neighbours overlap (Jaccard similairty).

Note: The figure reports pairwise Jaccard similarities between physicists' sets of first-order contacts in the undirected and unweighted correspondence network for 1900–1925. Jaccard similarity is defined as the number of contacts shared by two physicists divided by the total number of unique contacts belonging to either physicist's network. Darker blue indicates greater neighborhood overlap. The matrix is symmetric, and diagonal values are omitted because each physicist's contact set is perfectly similar to itself.

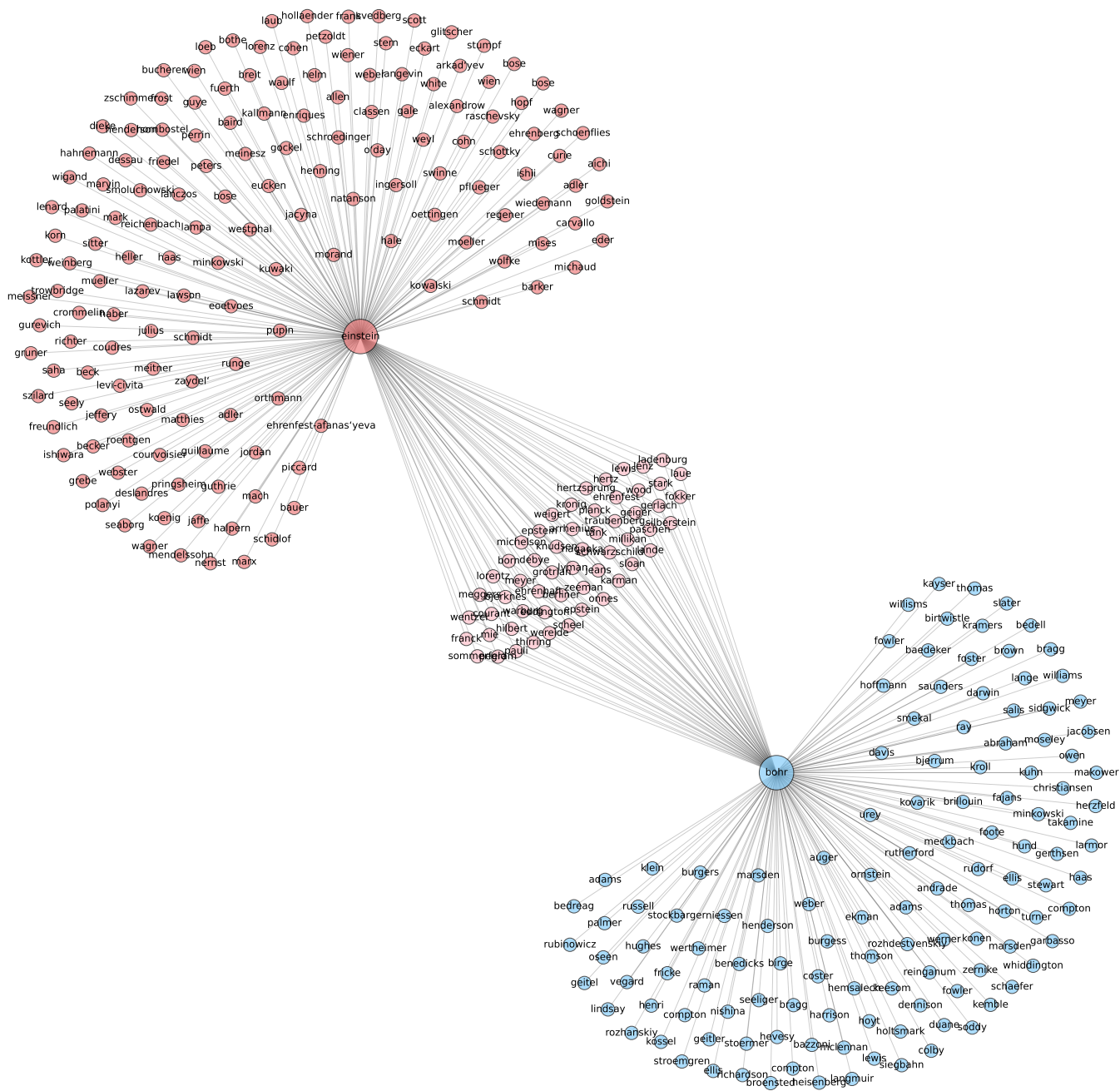


FIGURE 7. Social graph of core and their first neighbors.

Note: Undirected network. Years 1900-06, 1907-13, 1914-21, and 1922-25 are bunched together. Spring layout: $k=0.35$, iterations=200. Graph built using Python networkx library. Connections between the first neighbors (the second neighbors) are deliberately excluded to make accent on the overlap between the correspondents of Bohr and Einstein.

TABLE 4. Network coverage from Bohr and Einstein across correspondence periods

Period	Core physicist	All nodes	Reached nodes	Share reached	Unreached nodes
1900–1906	Bohr	609	583	0.957	26
1900–1906	Einstein	609	588	0.966	21
1907–1913	Bohr	836	821	0.982	15
1907–1913	Einstein	836	820	0.981	16
1914–1921	Bohr	989	980	0.991	9
1914–1921	Einstein	989	980	0.991	9
1922–1925	Bohr	954	934	0.979	20
1922–1925	Einstein	954	934	0.979	20
1914–1925	Bohr	1,325	1,312	0.990	13
1914–1925	Einstein	1,325	1,312	0.990	13
1900–1925	Bohr	1,626	1,612	0.991	14
1900–1925	Einstein	1,626	1,612	0.991	14

Notes: Reached nodes are physicists connected to the indicated core physicist through the correspondence network within the respective period, including successive network neighbors up to the fifth order. Share reached is the number of reached nodes divided by the total number of nodes present in the corresponding period. The network is treated as undirected.

31.1% of Bohr’s neighbors and 26.3% of Einstein’s neighbors.

In the Appendix, Figures ??, ??, ??, and ?? show how the correspondence network in the physics profession in the years approximately 1900 to 1925 was formed primarily around two major hubs: Niels Bohr and Albert Einstein. The overlap in the number of connections between the two hubs in the bunched years network is around 30%, which means that around 30% of active physicists were corresponding with both of the giants.

The Figures ?? shows the cumulative years evolution of the correspondence networks of physicists (e.g., for years 1914–21, the figure depicts cumulatively edges from years 1900–06, 1907–13, and 1914–21).

TABLE 5. Direct-neighborhood comparison of Bohr and Einstein, 1900–1925

Network measure	Value
Bohr direct neighbors	181
Einstein direct neighbors	214
Shared direct neighbors	56
Bohr-only direct neighbors	125
Einstein-only direct neighbors	158
Jaccard similarity	0.165

Notes: Direct neighbors are physicists connected by at least one correspondence link to the respective scientist. Shared neighbors are directly connected to both Bohr and Einstein. Bohr-only and Einstein-only neighbors are directly connected to only one of the two scientists. Jaccard similarity is defined as the number of shared neighbors divided by the number of physicists in the union of the two direct neighborhoods; it ranges from 0 to 1. The network is treated as undirected.

TABLE 6. Bohr’s strongest direct correspondence connections, 1900–1925

Physicist	ID	Specialty	Country	Letters
Paul Sigismondovich Ehrenfest	1289	QG	AU UR NE	89
James Franck	1536	QG	GE US	88
Wolfgang Pauli Jr.	3726	QG	AU SZ US	78
Gyorg von Hevesy	2133	PC	HU DK GE	77
Oskar Benjamin Klein	2629	QG	SD	67
Owen Willans Richardson	4053	QG	GB US	62
Ernest Rutherford	4226	NU	CA GB	60
Karl Friedrich F. C. Scheel	4316	DS	GE	53
Adalbert Wojciech Rubinowicz	4189	QG	PO	49
Alfred Lande	2839	QG	GE US	43
Hendrik Anthony Kramers	2736	QG	DK NE	41
Carl Wilhelm Oseen	3657	QG	SD	38
John Cunningham McLennan	3259	NU	CA GB	30
Hendrik Antoon Lorentz	3040	QG	NE	29
Alfred Fowler	1516	AP	GB	29
Max Karl Ernst Ludwig Planck	3830	QG	GE	26
Sophus Th. H. Weber	5263	TT	NE	22
Theodor L. Lyman	3082	TT	US	22
Ralph Howard Fowler	1521	QG	GB	19
Werner Heisenberg	2075	QG	GE	16

Notes: The table reports Bohr’s twenty direct neighbors with the largest total number of letters exchanged over 1900–1925. Letters are aggregated across all correspondence periods and across both directions of communication. Specialty and country codes follow the classifications used in the physicist-level data.

TABLE 7. Einstein’s strongest direct correspondence connections, 1900–1925

Physicist	ID	Specialty	Country	Letters
Paul Sigismondovich Ehrenfest	1289	QG	AU UR NE	267
Hendrik Antoon Lorentz	3040	QG	NE	173
Arnold Johannes Wilhelm Sommerfeld	4670	QG	GE	83
Erwin Finlay Freundlich	1567	AP	GE	79
Hermann Weyl	5330	MA	SZ US	76
Jakob Johann Laub	2880	QG	AR GE	67
Max Karl Ernst Ludwig Planck	3830	QG	GE	51
Wilhelm Carl Werner Otto F. F. Wien	5366	QG	GE	42
Willem de Sitter	4572	AP	NE	41
Fritz Haber	1939	PC	GE	40
David Hilbert	2151	MA	GE	38
Max Theodor Felix von Laue	2882	QG	GE	38
Paul Langevin	2852	QG	FR	37
George Ellery Hale	1957	AP	US	34
Gustav Mie	3343	QG	GE	30
Max Born	548	QG	GE	27
Robert William Lawson	2893	NU	GB	27
Arnold Berliner	408	DS	GE	25
Tullio Levi-Civita	2944	MA	IT	23
Robert Andrews Millikan	3365	QG	US	22

Notes: The table reports Einstein’s twenty direct neighbors with the largest total number of letters exchanged over 1900–1925. Letters are aggregated across all correspondence periods and across both directions of communication. Specialty and country codes follow the classifications used in the physicist-level data.

TABLE 8. Physicists in the Bohr and Einstein correspondence neighborhoods, 1900–1925

Network order	Mutually exclusive categories			Group totals		Unique
	Bohr only	Shared	Einstein only	All Bohr	All Einstein	total
First order	125	56	158	181	214	339
Second order	344	751	269	1,095	1,020	1,364
Third order	152	168	193	320	361	513
Fourth order	6	8	7	14	15	21
Fifth order	0	1	0	1	1	1

Notes: Network order is measured separately from Bohr and Einstein in the undirected historical correspondence network. “Shared” denotes physicists appearing in both leaders’ neighborhood sets at the given order. “Bohr only” and “Einstein only” exclude these shared physicists. All Bohr equals Bohr only plus shared; all Einstein equals Einstein only plus shared. The unique total is the union of the Bohr and Einstein sets within each row. Because distance is measured separately from each leader, a physicist may appear at different orders from Bohr and Einstein; consequently, the unique totals should not be summed across rows.

TABLE 9. Physicists in the Bohr and Einstein correspondence neighborhoods, 1900–1925

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Notes: Network order is measured separately from Bohr and Einstein in the undirected historical correspondence network. “Shared” denotes physicists appearing in both leaders’ neighborhood sets at the given order. “Bohr only” and “Einstein only” exclude these shared physicists. All Bohr equals Bohr only plus shared; all Einstein equals Einstein only plus shared. The unique total is the union of the Bohr and Einstein sets within each row. Because distance is measured separately from each leader, a physicist may appear at different orders from Bohr and Einstein; consequently, the unique totals should not be summed across rows.

TABLE 10. Jaccard similarity between physicists’ first-order correspondence neighborhoods, 1900–1925

Physicist pair	Jaccard similarity
Bohr–Einstein	0.1711
Bohr–Lyman	0.1222
Bohr–Stark	0.1636
Bohr–Millikan	0.1184
Bohr–Sommerfeld	0.2066
Bohr–Lorentz	0.1416
Bohr–Rutherford	0.1383
Bohr–Hale	0.1034
Bohr–Meggers	0.0992
Einstein–Lyman	0.0505
Einstein–Stark	0.1820
Einstein–Millikan	0.0812
Einstein–Sommerfeld	0.1765
Einstein–Lorentz	0.1955
Einstein–Rutherford	0.1121
Einstein–Hale	0.0969
Einstein–Meggers	0.0629
Lyman–Stark	0.0737
Lyman–Millikan	0.2063
Lyman–Sommerfeld	0.0731
Lyman–Lorentz	0.0821
Lyman–Rutherford	0.0914
Lyman–Hale	0.1563
Lyman–Meggers	0.1271
Stark–Millikan	0.0859
Stark–Sommerfeld	0.1778
Stark–Lorentz	0.2134
Stark–Rutherford	0.1776
Stark–Hale	0.1172
Stark–Meggers	0.0641
Millikan–Sommerfeld	0.1140
Millikan–Lorentz	0.1042
Millikan–Rutherford	0.1458
Millikan–Hale	0.2285
Millikan–Meggers	0.1379
Sommerfeld–Lorentz	0.2136
Sommerfeld–Rutherford	0.1241
Sommerfeld–Hale	0.1053
Sommerfeld–Meggers	0.0913
Lorentz–Rutherford	0.1774
Lorentz–Hale	0.1467
Lorentz–Meggers	0.0598
Rutherford–Hale	0.1213
Rutherford–Meggers	0.0734
Hale–Meggers	0.1548

Notes: Each neighborhood contains the unique first-order contacts of the respective physicist in the undirected and unweighted historical correspondence network. For physicists i and j , Jaccard similarity is calculated as $J_{ij} = |N_i \cap N_j| / |N_i \cup N_j|$, where N_i and N_j denote their respective sets of direct contacts. Higher values indicate greater overlap between the two correspondence neighborhoods.

5. Identification Strategy and Empirical Design

The empirical analysis asks whether scientists with pre-existing access to the correspondence networks of the founders of quantum mechanics experienced a different change in research output after the breakthrough of 1925 than otherwise comparable scientists without a direct link. The identifying variation is therefore not the arrival of quantum mechanics by itself: 1925 was a common shock to the physics profession. Instead, the design compares the differential response of scientists who entered the breakthrough with different degrees of pre-existing social access to the scientists developing the new theory.

5.1. Treatment, comparison group, and estimand

Let D_i indicate whether scientist i exchanged at least one letter, in either direction, with at least one of the eight founders of quantum mechanics—Niels Bohr, Max Born, Louis de Broglie, Paul Dirac, Werner Heisenberg, Pascual Jordan, Wolfgang Pauli, and Erwin Schrödinger—before the breakthrough. Thus,

$$(1) \quad D_i = \mathbf{1}\{\text{scientist } i \text{ had a direct correspondence link to a founder before 1925}\}.$$

The baseline definition treats a single letter as sufficient and does not use direction or volume. It is consequently a measure of direct network access rather than of the intensity or scientific content of the relationship. Scientists with $D_i = 0$ form the comparison group. Treatment status is fixed over the analysis window, while the post indicator is $Post_t = \mathbf{1}\{t \geq 1925\}$.

The archival source groups part of the correspondence into a 1922–1925 interval. Unless exact dates from the Archive for History of Quantum Physics or another source establish that a link was formed by the end of 1924, this interval cannot cleanly separate pre-existing access from contact initiated during 1925. I therefore interpret the baseline estimates as effects conditional on the recorded timing of correspondence. A strict causal interpretation requires either an end-of-1924 treatment measure or a conservative treatment measure using links observed through 1921.

The target estimand is the average differential post-1925 change among scientists who were already active at the scientific frontier before the breakthrough. In potential-outcome notation, for $t \geq 1925$ it is

$$(2) \quad ATT(t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid D_i = 1],$$

where $Y_{it}(1)$ is the outcome scientist i would produce in year t with pre-existing direct access to a founder and $Y_{it}(0)$ is the corresponding outcome without that direct access. Because unconnected scientists could learn through journals, conferences, colleagues, and indirect paths in the correspondence network, $ATT(t)$ is a relative advantage of direct access. It is not the total

effect of the quantum-mechanics revolution on physics.

The analysis uses an annual scientist panel for 1919–1932. The intended risk set consists of physicists who had begun publishing by the end of 1924 and were alive in 1925; the founders themselves are excluded. The current results reported below contain 18,802 scientist-year observations for 1,343 scientists. The sample-selection rules, including the treatment flag and the handling of years before entry or after death, must be applied in a single reproducible analysis file because coding inactive years as zero can otherwise confound a productivity response with changes in sample composition.

5.2. Outcomes and pre-treatment covariates

I study four annual outcomes. The extensive-margin outcome is an indicator for publishing at least one article in year t . The intensive-margin outcome is $\log(1 + \text{publications}_{it})$. To measure the reception of new work, I use $\log(1 + \text{citations}_{it}^5)$ and $\log(1 + \text{citations}_{it}^{25})$, where citations are summed over the first five or twenty-five years after publication for the articles that scientist i published in year t . This construction follows the use of publications and citation-weighted publications as complementary measures of scientific output in the economics-of-science literature (Waldinger 2012).

The propensity-score model uses characteristics determined before 1925: mean annual fractional authorship, the probability of publishing, short-horizon citations, field, country of work, birth year, and first publication year. Pre-treatment levels and slopes in the production outcomes are the most informative diagnostics of comparability. Long-horizon forward citations require a separate timing caveat: citations assigned to a pre-1925 publication can be received after 1925. Consequently, a lead coefficient for twenty-five-year citations is not a clean placebo for the absence of anticipation. I use publication outcomes to assess pre-trends and interpret the long-horizon citation event study as evidence on the later reception of papers from different publication cohorts.

5.3. Econometric specification

As a transparent benchmark, I estimate the following two-way fixed-effects difference-in-differences specification separately for each outcome:

$$(3) \quad Y_{it} = \alpha_i + \lambda_t + \beta (D_i \times \text{Post}_t) + \varepsilon_{it},$$

where α_i are scientist fixed effects and λ_t are publication-year fixed effects. Scientist fixed effects absorb time-invariant differences in ability, field, and career-long productivity; year fixed effects absorb shocks common to all sampled physicists. The coefficient β summarizes the average post-1925 change for connected scientists relative to the contemporaneous change for unconnected scientists. Standard errors are clustered by scientist.

The preferred dynamic analysis follows the conditional difference-in-differences framework of [Callaway and Sant’Anna \(2021\)](#). Let X_i collect the pre-treatment covariates and let $p(X_i) = \Pr(D_i = 1 \mid X_i)$ denote the propensity score. In the inverse-probability-weighted estimator, treated scientists receive unit weight and comparison scientists are reweighted by the odds $p(X_i)/[1 - p(X_i)]$, with the weights normalized in estimation. This makes the weighted distribution of observed pre-treatment characteristics in the comparison group resemble that of the connected group, provided that the two groups have adequate common support. The doubly robust estimator combines this propensity-score model with an outcome-regression model; under the regularity conditions in [Sant’Anna and Zhao \(2020\)](#), it remains consistent when either nuisance model is correctly specified. Double robustness does not, however, remove the need for conditional parallel trends, overlap, and valid treatment timing.

To display dynamics, I estimate event-time effects relative to 1924:

$$(4) \quad Y_{it} = \alpha_i + \lambda_t + \sum_{e \in \mathcal{E}, e \neq -1} \beta_e D_i \mathbf{1}\{t - 1925 = e\} + \varepsilon_{it},$$

where $e = -1$ is the omitted reference year. In the weighted implementation, β_e is reported as the event-time average treatment effect for the connected group. The pre-period coefficients summarize differential changes relative to 1924; the post-period coefficients trace the evolution of the connected-group advantage after the breakthrough. The event-study figures report wild-bootstrap confidence intervals clustered at the scientist level.

5.4. Identifying assumptions and interpretation

The central identifying assumption is conditional parallel trends. For every post-treatment year t ,

$$(5) \quad \mathbb{E}[Y_{it}(0) - Y_{i,1924}(0) \mid D_i = 1, X_i] = \mathbb{E}[Y_{it}(0) - Y_{i,1924}(0) \mid D_i = 0, X_i].$$

This states that, after conditioning on observed pre-1925 characteristics, connected and unconnected scientists would have experienced the same change in the outcome in the absence of the differential advantage supplied by direct founder access. Reweighting can improve balance in observed characteristics; it cannot “engineer” this counterfactual assumption or eliminate selection on unobserved time-varying determinants of productivity. The large unweighted differences in pre-1925 output documented below make overlap and weighted balance substantive diagnostics rather than mechanical preliminaries.

Three additional conditions matter. First, common support requires $0 < p(X_i) < 1$ for the treated population of interest; the distribution of propensity scores and the largest weights should therefore be reported. Second, no anticipation requires that scientists did not change their output before 1925 because they foresaw the breakthrough. The historical abruptness of

matrix mechanics supports this assumption, but the timing of the correspondence measure must also be strictly pre-treatment. Third, the conventional estimator assumes no consequential interference across scientists. That condition is demanding in a correspondence network: an unconnected scientist may be indirectly exposed through a connected colleague. Methods for spillovers show that ignoring such interference can combine direct and indirect effects in nontransparent ways (Vazquez-Bare 2023). Accordingly, the estimates below should be read as the conditional differential change associated with a direct founder link, relative to scientists without such a link, and not as a pure individual-level treatment effect free of network diffusion.

6. Empirical Results

6.1. Baseline differences and fixed-effects estimates

Connected scientists were already positively selected before 1925. In the current balanced panel, their pre-period probability of publishing was 0.428, compared with 0.295 among unconnected scientists. Their mean values of $\log(1 + \text{publications})$, $\log(1 + \text{five-year citations})$, and $\log(1 + \text{twenty-five-year citations})$ were also substantially higher. These level differences do not by themselves invalidate difference-in-differences, because scientist fixed effects remove permanent gaps. They do show why the analysis must compare changes, assess pre-treatment slopes, and demonstrate balance and overlap after weighting.

Table 11 reports the unweighted fixed-effects benchmark. After 1925, connected scientists were 5.4 percentage points more likely to publish in a given year relative to the change among unconnected scientists. The clustered standard error is 2.2 percentage points, implying an approximate 95 percent confidence interval of 1.1 to 9.7 percentage points. Relative to the connected group's pre-period mean of 0.428, the point estimate is equivalent to a 12.6 percent increase in the annual probability of publishing.

The coefficient for $\log(1 + \text{publications})$ is 0.042, with a standard error of 0.024. Its sign is positive, but its approximate 95 percent confidence interval includes zero. The citation estimates are larger and more precisely estimated: 0.123 log points for citations accumulated within five years and 0.131 log points for citations accumulated within twenty-five years. The latter corresponds to approximately a 14.0 percent increase in $1 + \text{citations}$, not mechanically to a 14.0 percent increase in the raw citation count. Taken together, the benchmark estimates indicate that the post-1925 difference is more pronounced for the reception of scientific work than for the number of papers alone.

6.2. Weighted event-study evidence

The inverse-probability-weighted event study for $\log(1 + \text{twenty-five-year citations})$ yields pre-1925 estimates centered near zero, while every displayed post-1925 estimate is positive. The post-treatment profile rises in the first years after the breakthrough, is largest around event years

TABLE 11. Pre-existing correspondence links and research output after 1925

	Dependent variable			
	Any publication	$\log(1 + \text{pubs})$	$\log(1 + \text{cites}_5)$	$\log(1 + \text{cites}_{25})$
$D_i \times \text{Post}_t$	0.054*** (0.022)	0.042*** (0.024)	0.123*** (0.041)	0.131*** (0.052)
Scientist fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Treated pre-period mean	0.428	0.441	0.475	0.668
Scientists	1,343	1,343	1,343	1,343
Observations	18,802	18,802	18,802	18,802
R^2	0.239	0.315	0.337	0.325
Adjusted R^2	0.180	0.262	0.286	0.273

Notes: The table reports estimates of equation (3) for 1919–1932. D_i equals one for a scientist with a recorded pre-existing correspondence link to at least one founder of quantum mechanics, and Post_t equals one from 1925 onward. Standard errors clustered by scientist are in parentheses. *** $p < 0.01$.

three and four, and remains positive through 1932. The doubly robust specification produces a similar pattern. This stability across IPW and doubly robust estimators indicates that the dynamic result is not driven solely by one choice of nuisance model.

Two qualifications are important. First, the confidence intervals are wide at several horizons, so the year-to-year movements should not be overinterpreted. The event study is most informative about a sustained positive post-1925 pattern rather than the exact timing of a peak. Second, because the twenty-five-year citation window extends beyond the publication year, its pre-period coefficients cannot establish the absence of anticipatory behavior. They can partly capture the post-1925 recognition of papers published before the breakthrough. The cleaner evidence on pre-treatment trends must come from publication incidence, publication counts, and fractional authorship.

6.3. Event Study Diff-in-Diffs with IPW Results

6.4. Field-specific evidence and placebo comparison

Restricting the sample to theoretical quantum studies leaves 8,008 scientist-year observations and 572 scientists, of whom 180 are connected and 392 are unconnected. In this more substantively proximate field, the pooled estimate for $\log(1 + \text{twenty-five-year citations})$ is approximately 0.13 log points, close to the full-sample benchmark. The doubly robust event-study coefficients are positive throughout the post-1925 period and are largest in the middle post-treatment years. This pattern is consistent with the proposed mechanism: pre-existing access to the founders should matter most for scientists working close to the new theory.

As a negative-control comparison, the corresponding event study for astrophysics and physical chemistry does not show a sustained positive post-1925 path. Its coefficients fluctuate around zero and are imprecisely estimated. The contrast with theoretical quantum studies is suggestive, although it is not a formal test of equality across fields and should not be interpreted as eliminating all field-specific confounding.

6.5. What the estimates establish

Across the fixed-effects benchmark and the weighted dynamic specifications, the available results show a positive post-1925 differential for scientists with direct correspondence links to the founders. The strongest evidence concerns the probability of publishing and the subsequent citation reception of their work. The persistence of the citation pattern in the theoretical-quantum subsample, together with the absence of a comparable pattern in the astrophysics and physical-chemistry comparison, is consistent with a role for pre-existing social access in the diffusion and productive use of the new ideas.

The evidence remains conditional rather than design-based. Founder links were not randomly assigned; connected scientists were more productive before 1925; the treatment measure

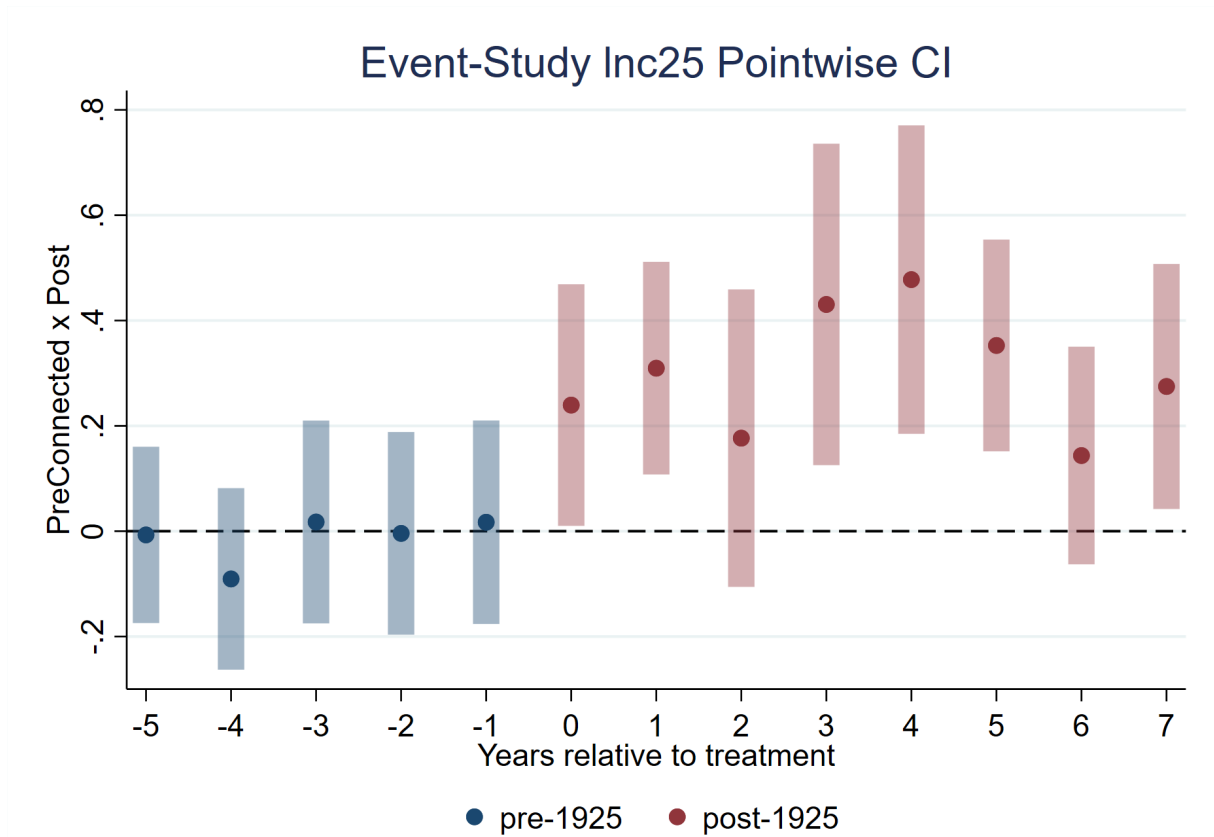


FIGURE 8. Inverse-probability-weighted event-study estimates for 25-year citations

Note: The figure reports event-time average treatment effects from an inverse-probability-weighted difference-in-differences estimator. The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the logarithm of one plus citation-weighted publications, with forward citations counted over the first 25 years after publication. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. Blue markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors; the dashed horizontal line denotes zero. The control group is reweighted using propensity scores based on pre-1925 mean publication share, probability of publishing, three- and five-year forward citations, scientific specialty, principal country of work, birth year, and first publication year. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

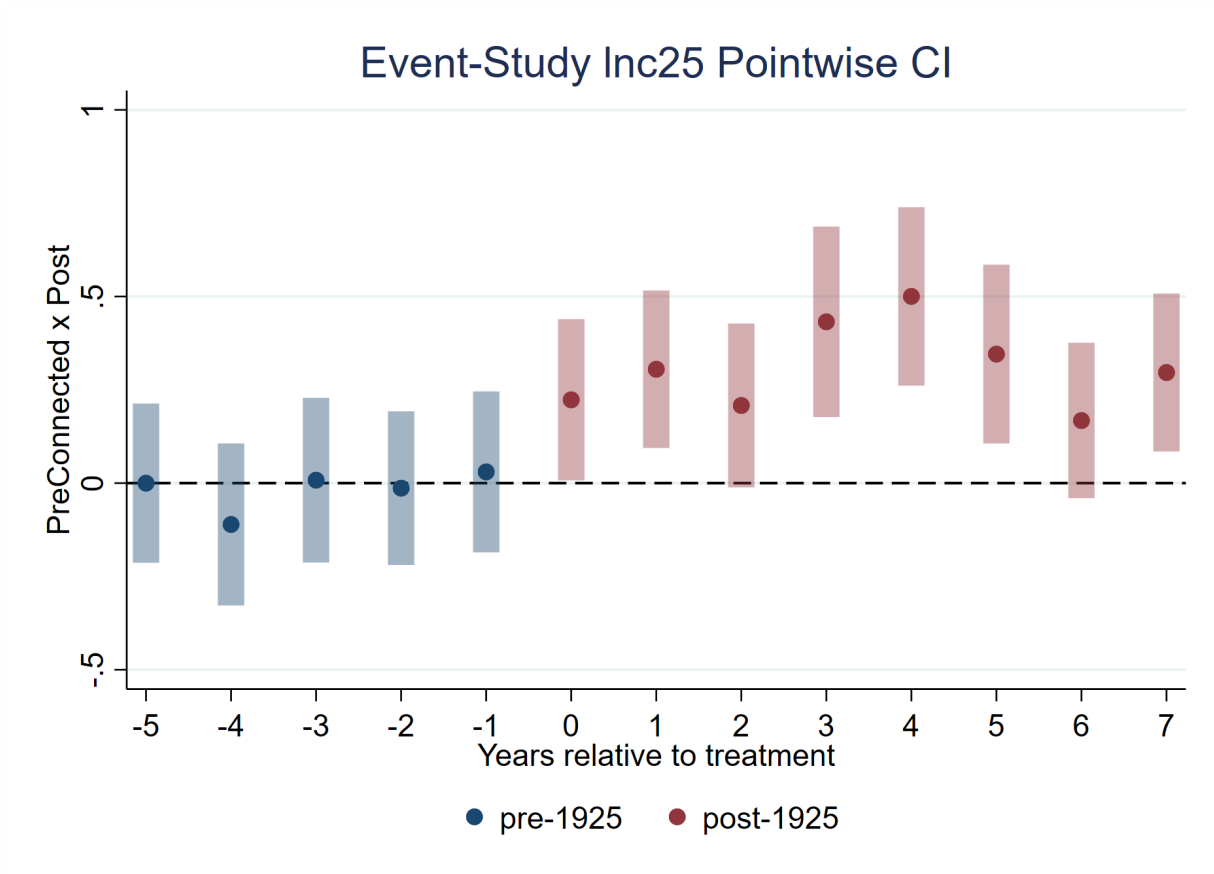


FIGURE 9. Doubly robust event-study estimates for 25-year citations

Note: The figure reports event-time average treatment effects from a doubly robust inverse-probability-weighted difference-in-differences estimator. Scientists preconnected through correspondence with at least one founder before 1925 form the treated group; scientists without such a connection form the control group. The outcome is the logarithm of one plus citation-weighted publications, with forward citations accumulated over 25 years after publication. Event time zero corresponds to 1925. Blue markers denote pre-treatment estimates, red markers denote post-treatment estimates, and the dashed horizontal line marks zero. Shaded bands are 95% pointwise confidence intervals constructed using wild-bootstrap standard errors. The estimator combines inverse-probability weights with an outcome-regression adjustment and uses pre-1925 publication share, publishing probability, three- and five-year forward citations, specialty, principal country of work, birth year, and first publication year as covariates. This doubly robust specification provides a robustness check on the baseline weighted estimates.

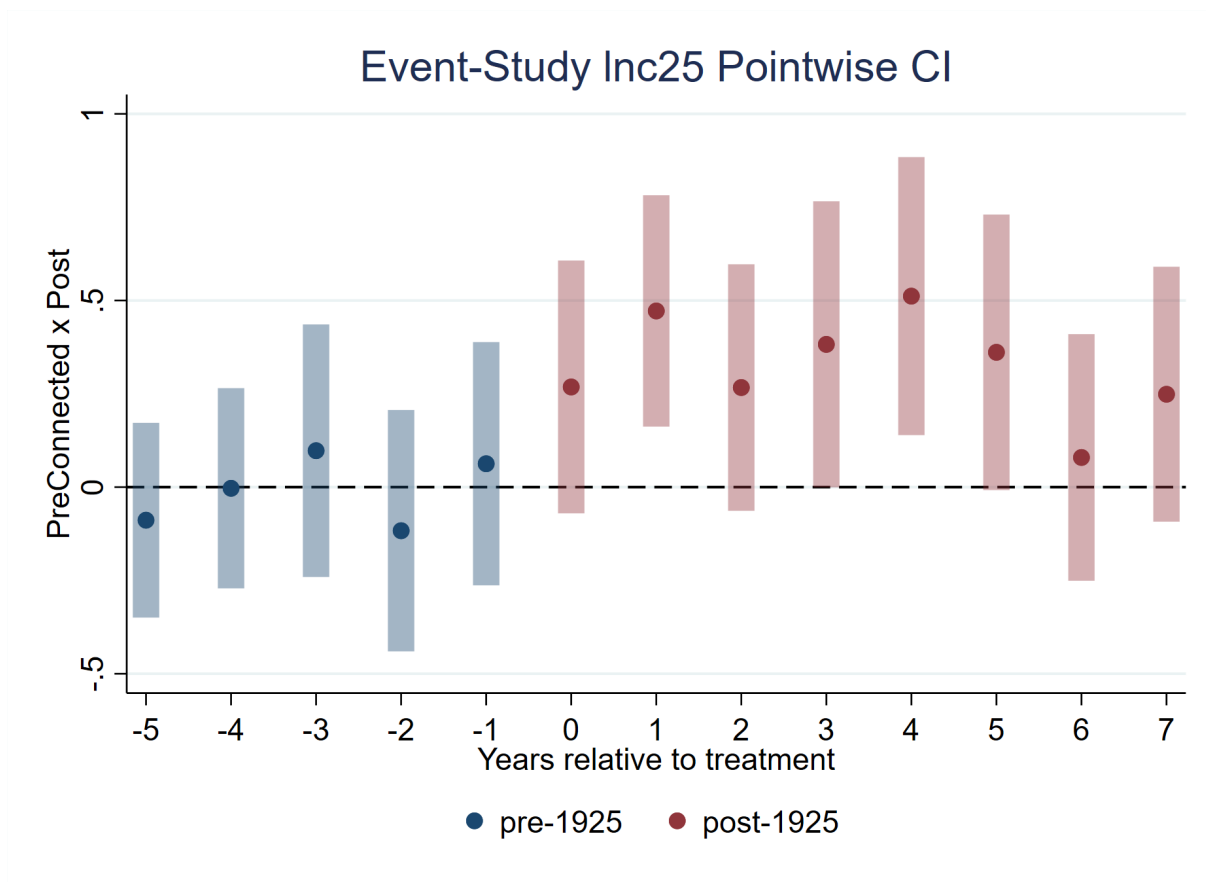


FIGURE 10. Doubly robust event-study estimates for theoretical quantum studies

Note: The figure repeats the doubly robust inverse-probability-weighted event study after restricting the sample to theoretical quantum studies. The restricted sample contains 8,008 scientist-year observations for 572 scientists, of whom 180 are treated and 392 are controls. Treatment is defined by a pre-1925 correspondence connection to at least one founder of quantum mechanics. The outcome is the logarithm of one plus citation-weighted publications, with citations measured over the first 25 years after publication, and event time zero denotes 1925. Blue markers show pre-treatment estimates and red markers show post-treatment estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors; the dashed horizontal line denotes zero. The estimator conditions on the same pre-discovery productivity and scientist characteristics as the full-sample specification. This restriction tests whether the estimated post-1925 citation response remains visible among scientists working most directly in the field affected by the breakthrough.

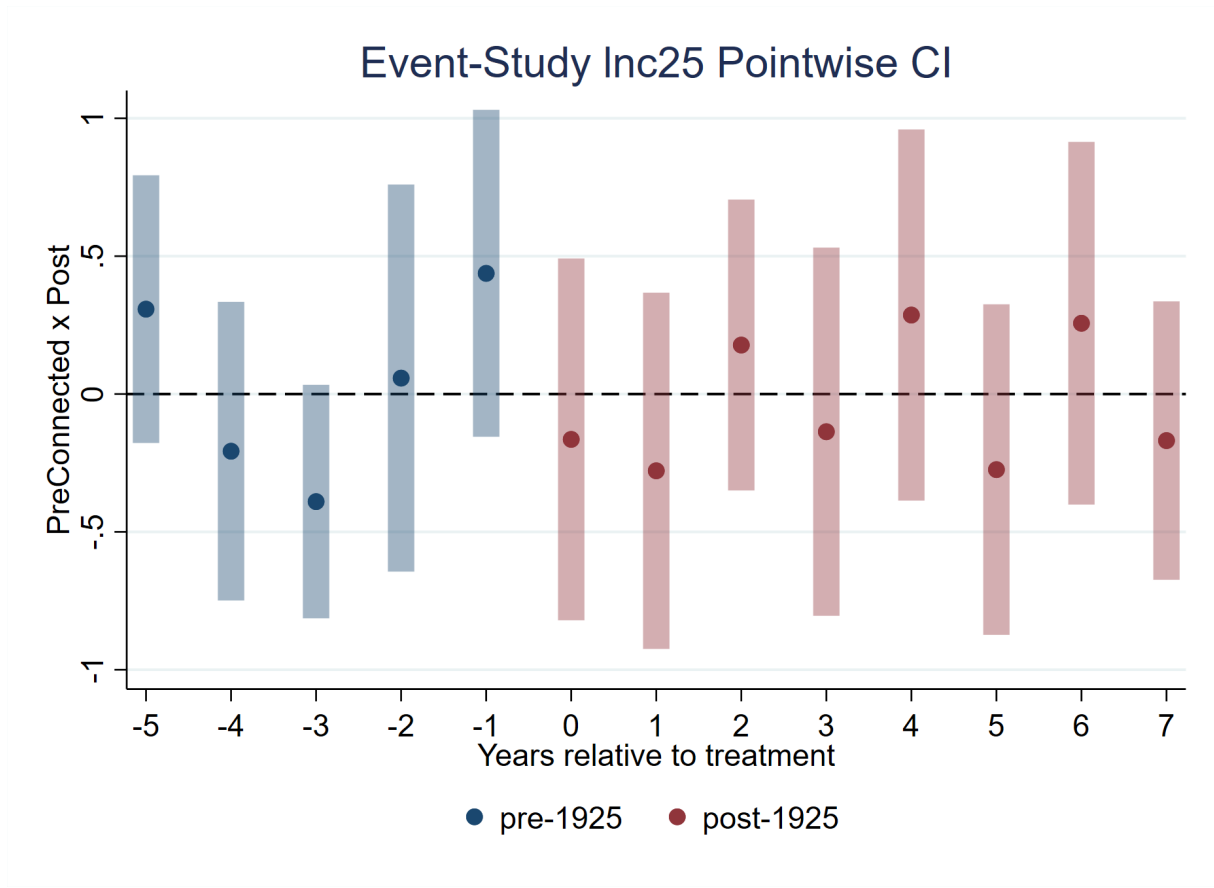


FIGURE 11. Doubly robust event-study estimates for astrophysics and physical chemistry

Note: The figure repeats the doubly robust inverse-probability-weighted event study for scientists classified in astrophysics or physical chemistry and excludes scientists classified in theoretical quantum studies. Treatment is defined by a pre-1925 correspondence connection to at least one founder of quantum mechanics. The outcome is the logarithm of one plus citation-weighted publications, using forward citations accumulated over 25 years after publication, and event time zero corresponds to 1925. Blue markers report pre-treatment estimates, red markers report post-treatment estimates, and the dashed horizontal line marks zero. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors. The estimator uses the same pre-1925 productivity measures and scientist characteristics as the baseline specification. The estimates are comparatively imprecise and fluctuate around zero before and after 1925, so this figure serves as a field-based comparison for the more concentrated response in theoretical quantum studies.

may include an interval that overlaps 1925; and ideas could diffuse to nominal controls through indirect network paths. IPW and doubly robust estimation address observed compositional differences but not selection on unobserved time-varying factors. For these reasons, the most defensible conclusion is that pre-existing direct correspondence with the founders predicts a larger post-1925 increase in scientific output and recognition among observably comparable physicists. Establishing a causal effect of personal communication more sharply requires strict pre-1925 dating of links, transparent common-support diagnostics, a consistently defined incumbent risk set, and an explicit treatment of network spillovers.

6.6. Diff-in-Diffs results accounting for the interference

This section will present the results with accounting for the interference in a simple and tractable way by introducing a measure of the Bohr-group exposure.

7. Conclusion

This paper examines whether pre-existing personal communication helped scientists respond to the quantum-mechanics revolution. I combine a newly digitized inventory of correspondence among physicists with publication and citation records, comparing scientists who corresponded with at least one founder before 1925 with scientists who lacked a direct link. The setting captures a central feature of scientific production: personal access to the creators of new knowledge may affect how productively others use it.

The results show a positive post-1925 differential for connected scientists. In the fixed-effects benchmark, their annual probability of publishing increased by 5.4 percentage points relative to the change among unconnected scientists. Their work received approximately 0.12–0.13 log points more citations over five- and twenty-five-year windows, while the publication-count estimate is positive but imprecise. Weighted and doubly robust event studies show a sustained positive citation differential. The pattern remains present in theoretical quantum studies, whereas astrophysics and physical chemistry show no comparable sustained response. These findings are consistent with prior access to the founders facilitating the production and recognition of research related to the new mechanics.

The evidence does not establish an unqualified causal effect. Scientists selected their correspondents, and connected scientists were more productive before the breakthrough. Reweighting improves comparability on observables but cannot remove selection on unobserved, changing factors. The timing of some links overlaps 1925, the letters' content is unknown, and indirect diffusion may expose scientists in the comparison group. Long citation windows also complicate pre-treatment coefficients. I therefore interpret the estimates as conditional differential changes associated with direct founder access, not as the total effect of the revolution or a pure effect of correspondence.

Further work can sharpen the design by dating links strictly before 1925, distinguishing sustained exchange from occasional contact, and documenting common support after weighting. Modelling indirect exposure would separate direct access from network spillovers, while cited-reference data could test whether connected scientists adopted the founders' ideas earlier. The central implication is that the productive use of a scientific breakthrough may depend not only on publication, but also on the social structure through which scientists communicate.

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Appendix A. Appendix title

A.1. Evolution of correspondence connections between physicists over time

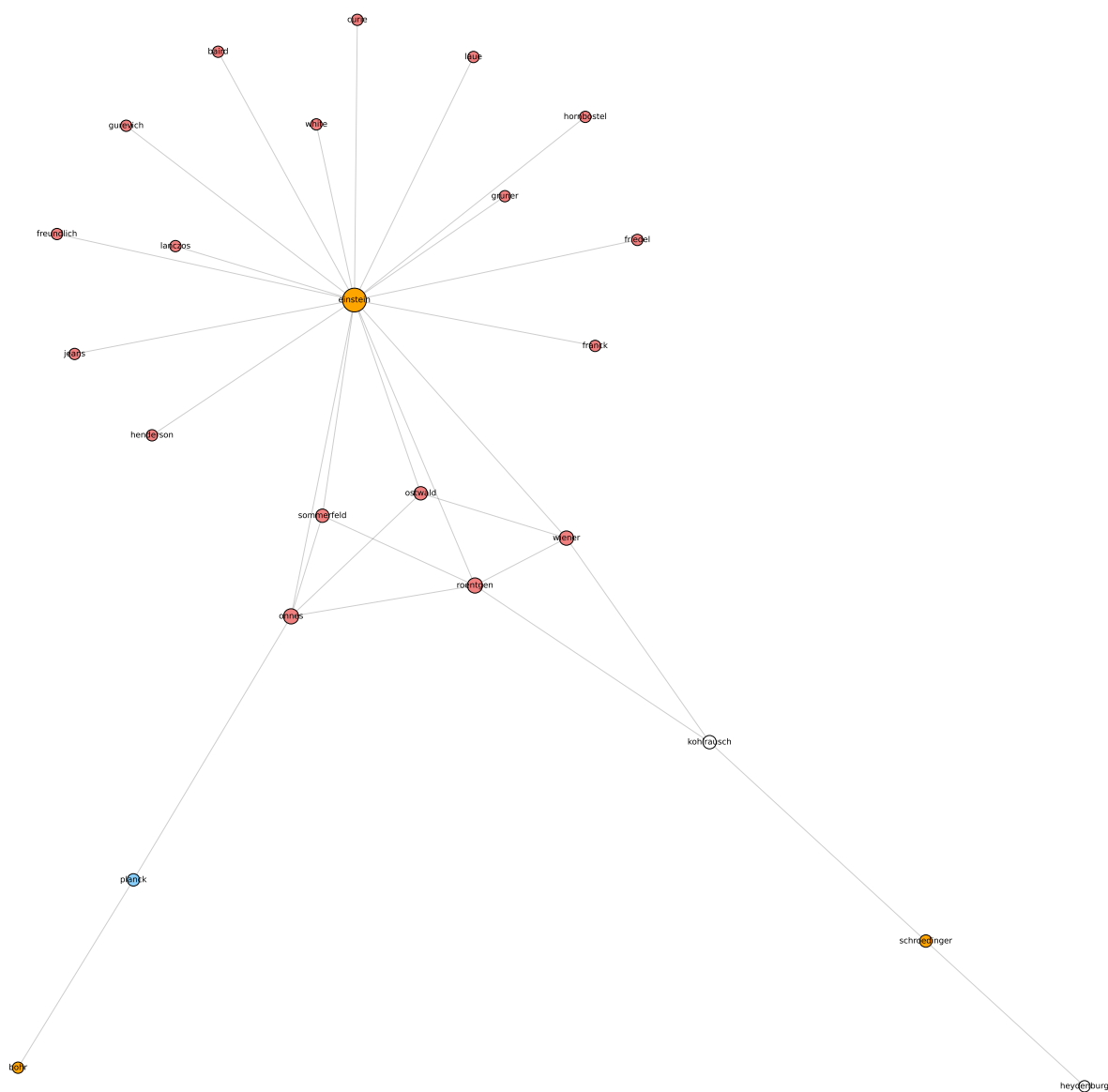


FIGURE A1. Correspondence network of core physicists and their first neighbors, 1900–1906.

Note: Undirected network.

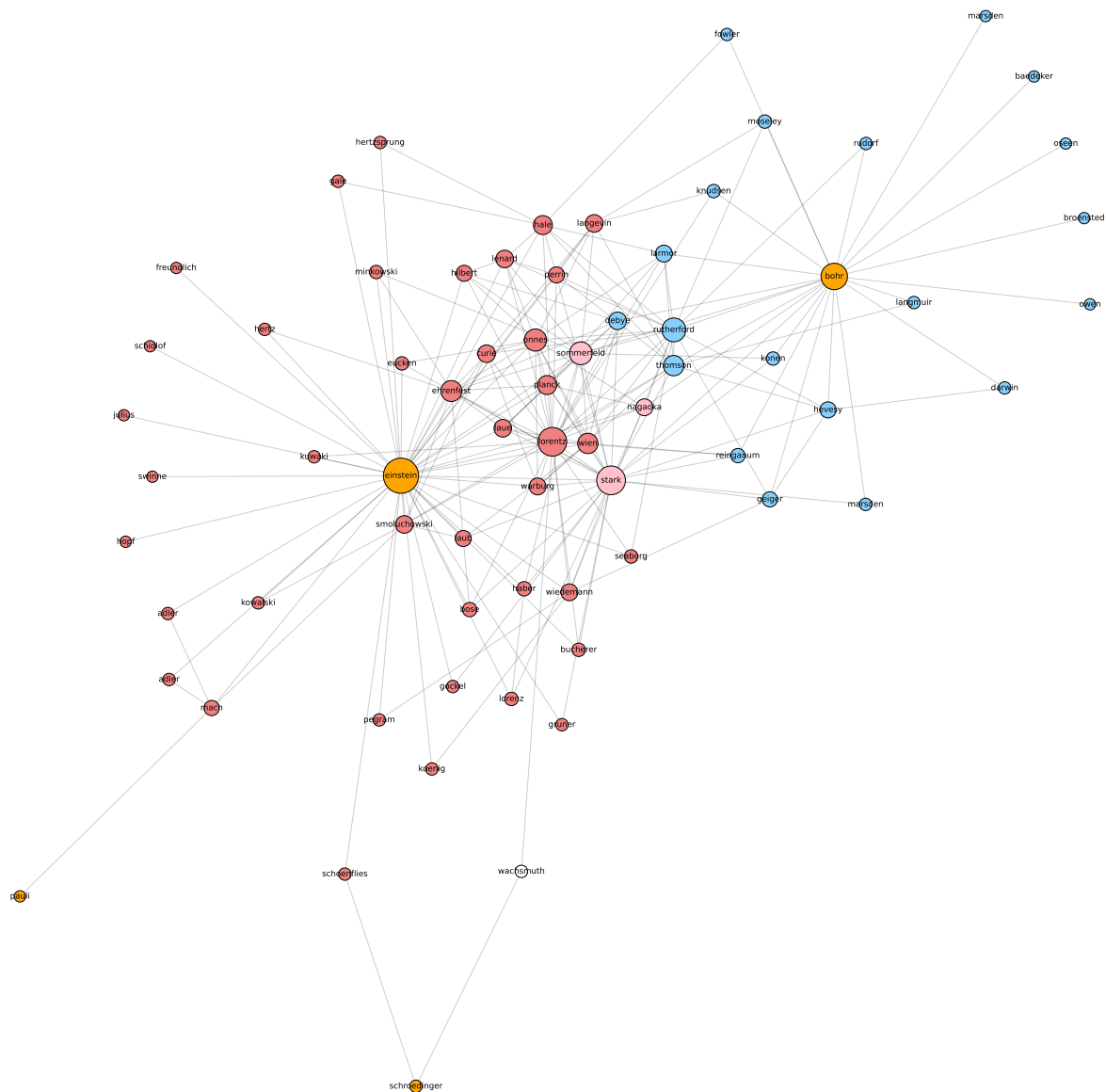
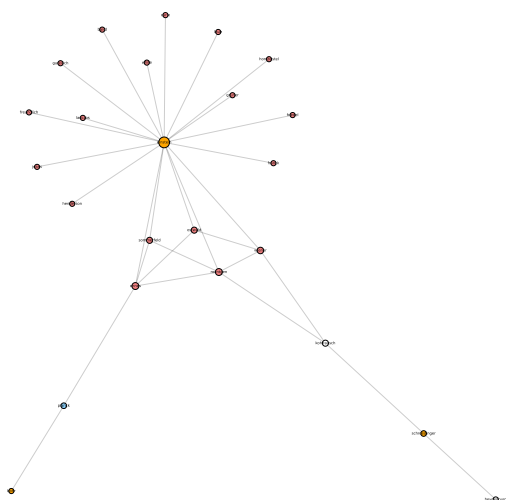
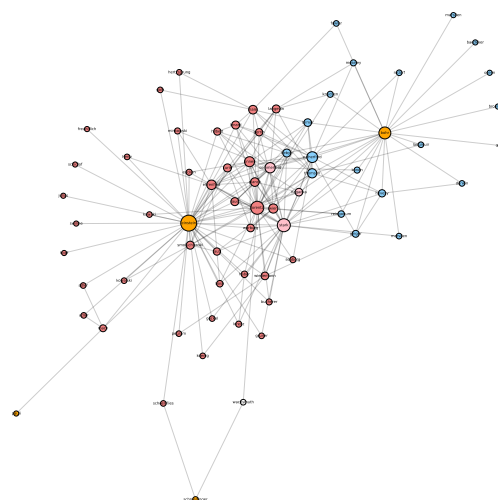


FIGURE A2. Correspondence network of core physicists and their first neighbors, 1907–1913.

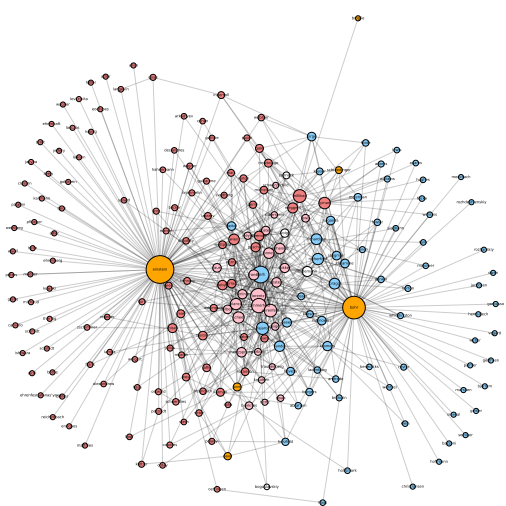
Note: Undirected network.



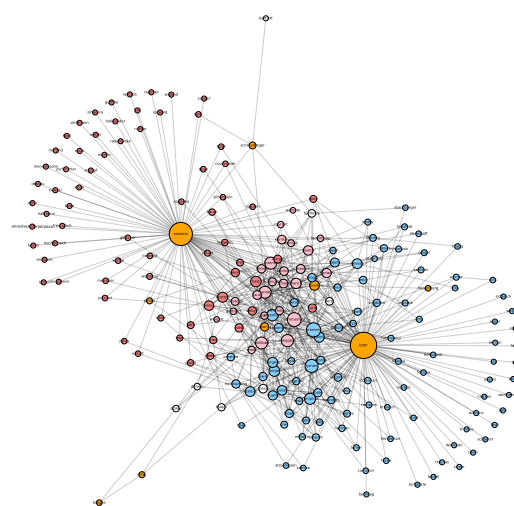
A. 1900-1906



B. 1907-1913



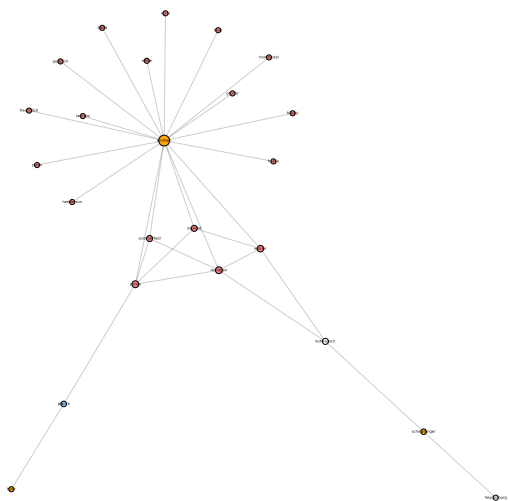
C. 1914-1921



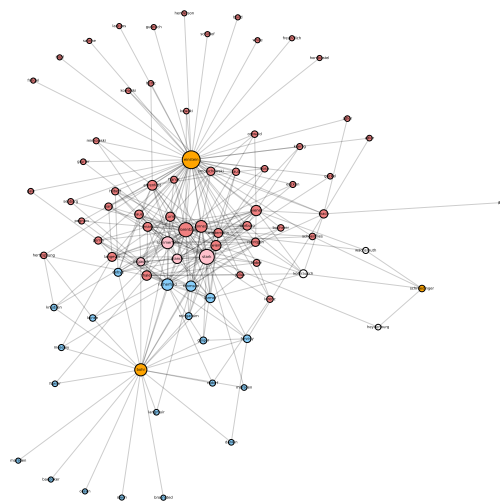
D. 1922-1925

FIGURE A5. Evolution of the correspondence network of core physicists and their first neighbors, 1900-1925.

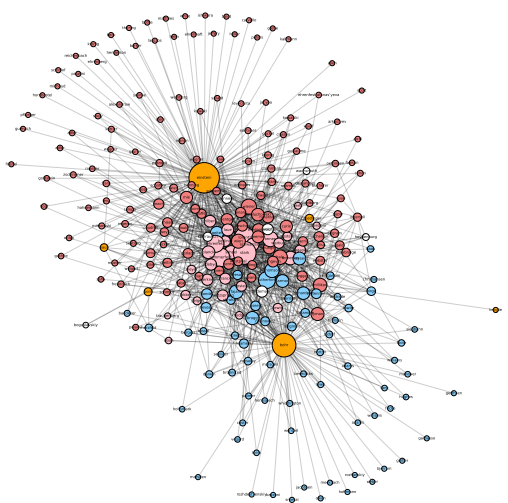
Note: Undirected networks.



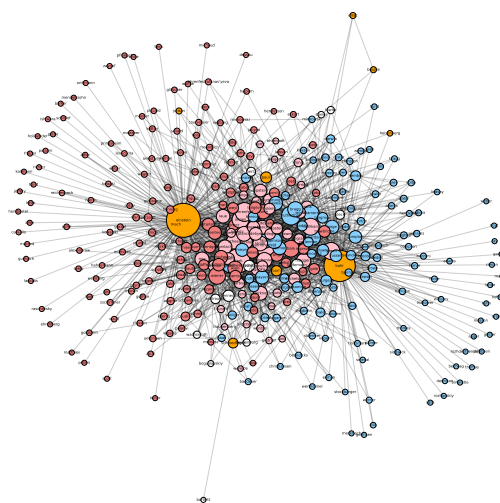
A. 1900–1906



B. 1900–1906, 1907–1913



C. 1900–1906, 1907–1913, 1914–1921



D. 1900–1906, 1907–1913, 1914–1921, 1922–1925

FIGURE A6. Cumulative-by-years evolution of the correspondence network of core physicists and their first neighbors, 1900–1925.

Note: Undirected networks. Cumulative means that for each given year period, the figure depicts edges from all the previous periods together with the given period.

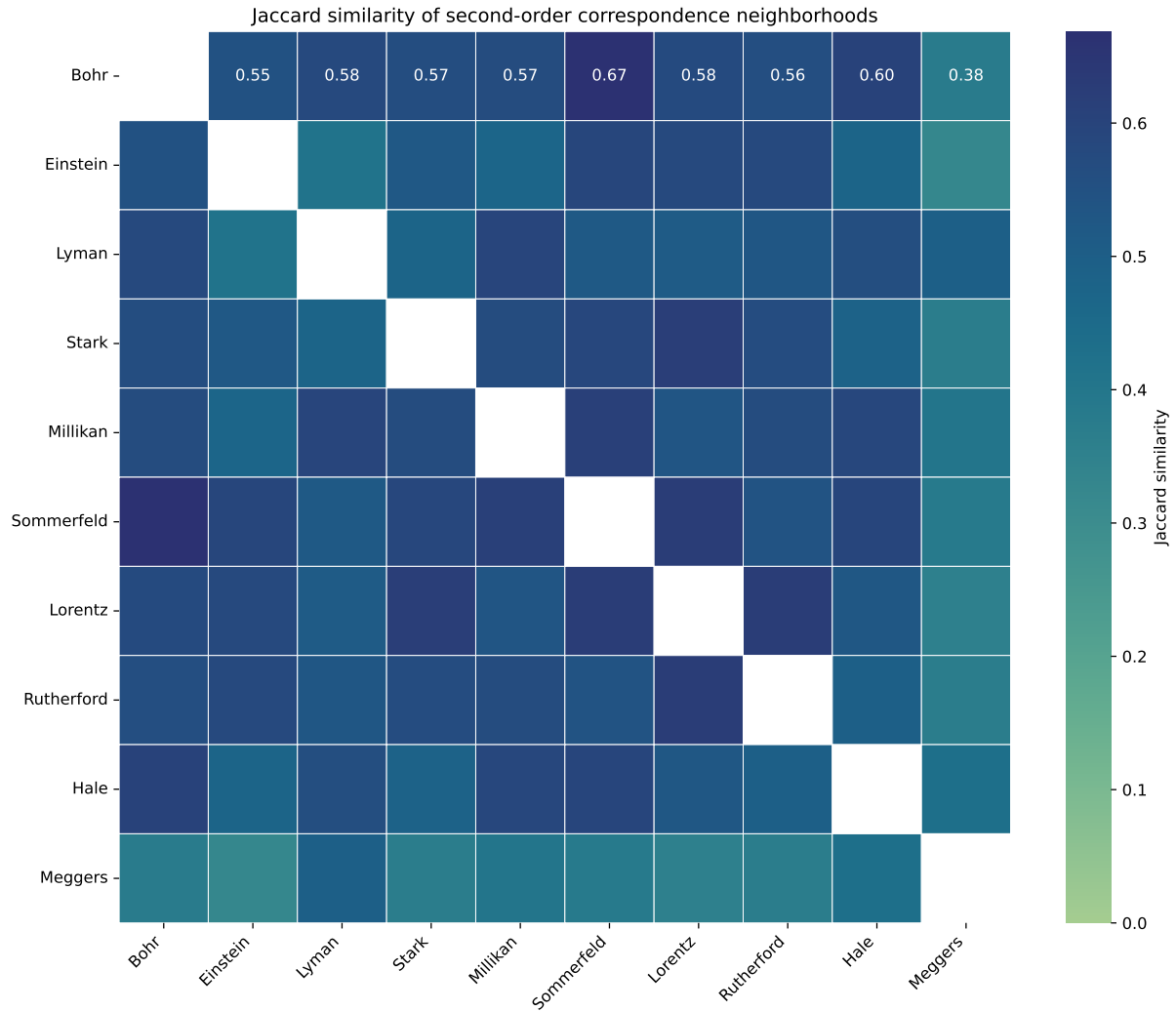


FIGURE A7. Heatmap of second neighbours overlap (Jaccard similairty).

Note: The figure reports pairwise Jaccard similarities between physicists' sets of second-order contacts in the undirected and unweighted correspondence network for 1900–1925. For each focal physicist, the second-order neighborhood contains contacts of their direct contacts, after removing the focal physicist and all first-order contacts. It therefore represents scientists reached at exactly two steps from the focal physicist. Jaccard similarity is the number of second-order contacts shared by two physicists divided by the total number of unique second-order contacts belonging to either physicist's network. Darker shading indicates greater neighborhood overlap. The matrix is symmetric, and diagonal values are omitted because each neighborhood is perfectly similar to itself.

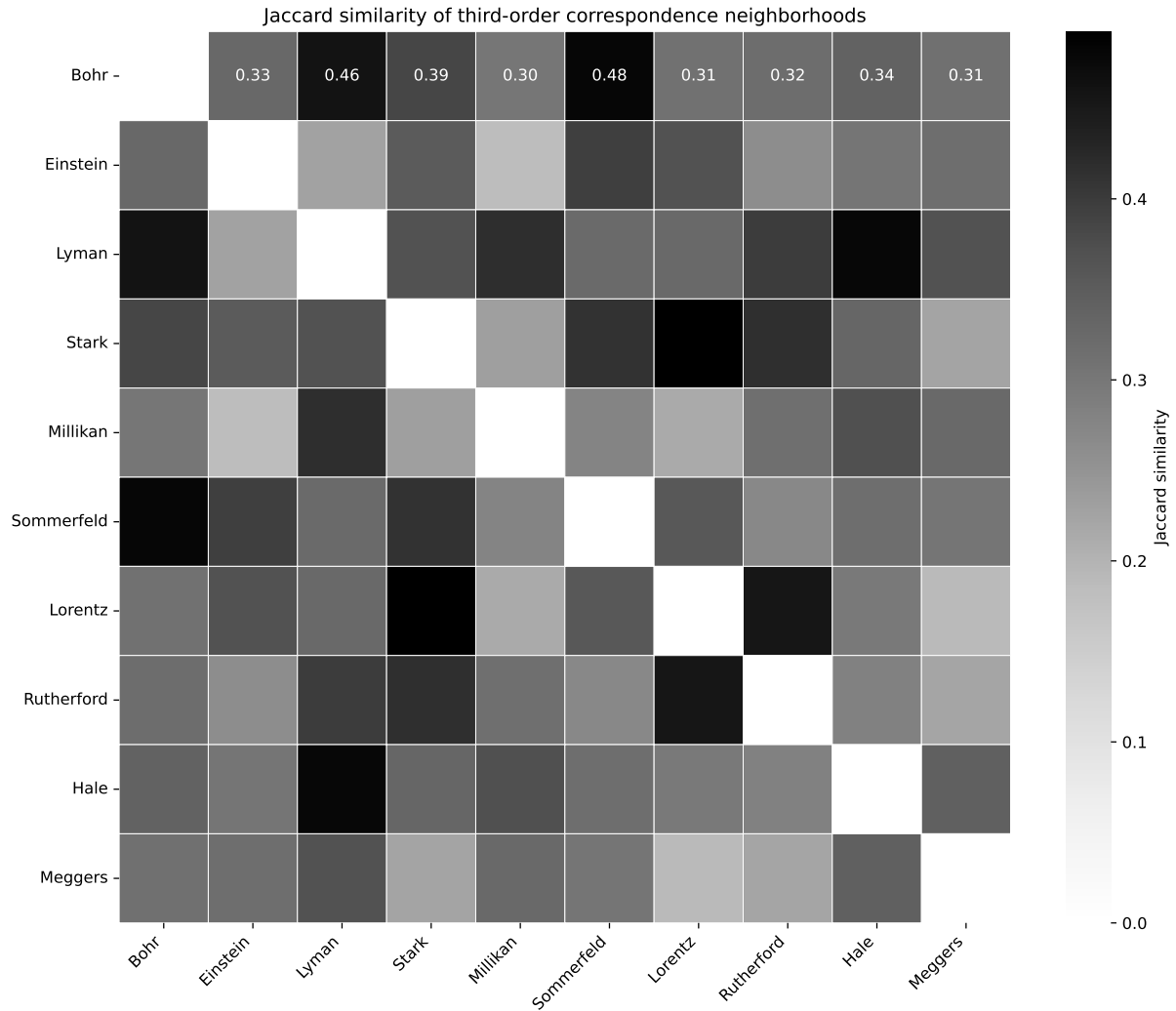


FIGURE A8. Heatmap of third neighbours overlap (Jaccard similarity).

Note: The figure reports pairwise Jaccard similarities between physicists' sets of third-order contacts in the undirected and unweighted correspondence network for 1900–1925. For each focal physicist, the third-order neighborhood contains contacts of their second-order contacts, after removing the focal physicist and all first- and second-order contacts. It therefore represents scientists reached at exactly three steps from the focal physicist. Jaccard similarity is the number of third-order contacts shared by two physicists divided by the total number of unique third-order contacts belonging to either physicist's network. Darker shading indicates greater neighborhood overlap. The matrix is symmetric, and diagonal values are omitted because each neighborhood is perfectly similar to itself.

